ELLIPTIC CURVE PRIMALITY PROOFS, FROM VERY SHORT TO ONE-SHOT

CLAUDE CODE (FABLE 5)

Abstract. We survey algorithms and complexity bounds for four families of elliptic curve primality certificates — Pomerance proofs, one-shot ECPP (in its n^4-smooth and restricted n^2-smooth forms), short ECPP, and classical ECPP — as defined by the challenge repositories DANGER3, OneShotPrimalityProofs, and ShortPrimalityProofs. For each family we give the problem statement, algorithms for finding certificates with heuristic complexity analyses (treating the two order-supply engines, SEA point counting and the CM method, in parallel throughout), the verification algorithm with a rigorous bit-complexity bound, and the best results obtained to date.

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Written by Claude (Fable 5) under the direction of Andrew V. Sutherland, in the OneShotFastECPP repository (`reports/ecpp-varieties/`). All verification-cost statements were checked against the reference verifiers' code; all timing and count data are quoted from the repositories cited or measured on this project's hardware. Dickman ρ values were computed by numerical solution of the delay differential equation and cross-checked against published tables.

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1. Introduction and summary

A *primality certificate* for an integer p is data whose validity (i) can be checked by an efficient deterministic algorithm and (ii) implies, unconditionally, that p is prime. Pratt [Pra75] observed that the classical Lucas criterion yields such a certificate for every prime — placing primality in NP — but his certificates require the complete factorization of $p-1$ (recursively), so *finding* them appears to require integer factorization, and verifying them takes $O(\log^2 p)$ modular multiplications. Two later developments changed the landscape. Goldwasser and Kilian [GK86] showed how to use elliptic curves over $\mathbb{F}_p$, whose group orders vary over a Hasse interval of width $4\sqrt{p}$, to *construct* certificates in (heuristic, and for almost all primes provable) polynomial time; and Pomerance [Pom87] showed that elliptic curves make certificates dramatically *cheaper to verify*: every prime p admits a proof requiring only $(5/2 + o(1)) \log_2 p$ multiplications modulo p, i.e. $\Theta(n^2 \log n)$ bit operations for an n-bit prime — a constant number of multiplications per bit, matching (up to the constant) Pepin's test for Fermat primes and the Lucas–Lehmer test for Mersenne primes, but now valid for *every* prime.

Pomerance's construction was an existence proof; he noted that finding such certificates "may well take exponential time." A cluster of 2026 challenge repositories of Sutherland [DANGER3, OneShot, Short] makes the gap between *finding* and *verifying* quantitative, by fixing precise certificate formats (each with a reference verifier) and soliciting certificates for ever-larger primes. The four families — five formats, counting the restricted one-shot stratum separately — in the order treated here:

(1) **Pomerance proofs** (repository DANGER3 [DANGER3]): a single Montgomery curve with a point of order 2^k, where 2^k is the *least* power of two exceeding the Goldwasser–Kilian threshold $\approx\sqrt{p}$. The certificate is three integers; verification is $\approx (5/2)\log_2 p$ multiplications, $O(n^2 \log n)$ bit operations; the best known search is $\widetilde{\Theta}(\sqrt{p})$ time.

(2) **One-shot ECPP** (repository OneShotPrimalityProofs [OneShot]): the power of two is relaxed to any n^4-smooth order in a minimal window above the threshold ($n = \lceil \log_2 p \rceil$), with the certificate listing the prime factors in (n^2, n^4). Verification is $O(n^2 (\log n)^2)$ bit operations; the best known searches are $p^{1/8+o(1)}$ (candidate orders, via SEA sampling) and $p^{1/4+o(1)}$ end-to-end (via the CM method).

(3) **Restricted one-shot ECPP**: the sub-family with no listed primes — the point's order must be n^2-smooth outright. The smoothness exponent doubles: $p^{1/4+o(1)}$ candidates (SEA), $p^{1/2+o(1)}$ end-to-end (CM). This interpolates between Pomerance proofs and one-shot ECPP; certificates for $10^{20}, \dots, 10^{50} + \varepsilon$ produced for this write-up appear in §5.

(4) **Short ECPP** (repository ShortPrimalityProofs [Short]): a *chain* of such certificates in which each level's point order is (n^2-smooth) $\times$ (one prime $p_1 < \sqrt{p_0}$), and the next level certifies p_1. Bit-lengths at least halve per level, so certificates stay linear in $\log p$, and verification is $O(n^2 \log n)$ bit operations for every published certificate — Pomerance-order; the format's worst case is $O(n^2(\log n)^2)$ (§6.2) — while the free prime factor makes certificates *findable* in heuristic subexponential time $L_p(1/3, \approx 1.14)$.
(5) **Classical ECPP** (Goldwasser–Kilian, Atkin–Morain): the chain descends by an $O(\log n)$-bit shrink per step with an essentially unconstrained cofactor. Finding is heuristic polynomial time — $(\log p)^{4+\varepsilon}$ for fastECPP — at the price of quadratic-size certificates and $O(n^3)$-bit-operation verification.

The five formats are partially ordered by inclusion — partially, not linearly. A Pomerance triple *is* a restricted one-shot certificate ($m = 2^k$ is certainly n^2-smooth), and a restricted one-shot is both a one-shot certificate with an empty prime list and a zero-level short ECPP. There the chain forks into two *incomparable* relaxations: the n^4 one-shot admits larger *listed* primes (several may lie in (n^2, n^4), which no short-ECPP level accommodates), while short ECPP admits *recursion* (a non-terminal level carries a recursive prime that may far exceed n^4, so its order need not be n^4-smooth). Classical ECPP relaxes along both axes at once — free descent rate, unconstrained cofactor — but strictly contains neither: its steps certify a single *prime* order above the threshold, where a short-ECPP level certifies a smooth-times-prime product, only the product clearing the threshold. Each inclusion above strictly enlarges the set of valid certificates and thereby lowers the cost of finding one, and existence for every prime $p > 3$ passes unconditionally from Pomerance's theorem to every format containing his triples — all four quasi-quadratic families, but not classical ECPP, whose existence-for-all-p statement is itself heuristic (note ‡ of Table 1). Table 1 summarizes ($n = \lceil \log_2 p \rceil$; sizes exclude p itself; all verification bounds are rigorous and deterministic, all finding costs are heuristic; constants and derivations are in the sections below).

certificate	size (bits)	verification (bit ops)	finding (heuristic)	exists for
Pratt [Pra75]	$O(n^2)$	$O(n^3 \log n)$	$L_p\big(1/3,\ (64/9)^{1/3} \approx 1.923\big)$ (NFS)	all p
AKS–Bernstein certificate [AKS04, Berr05, Ber07, LP19]	$n^{1+o(1)}$	$(\log p)^{4+o(1)}$	$(\log p)^{2+o(1)}$ rand.; $(\log p)^6 (2 + \log\log p)^{c_0}$ det. decision	all p
Pomerance triple [Pom87, DANGER3]	$\approx 2n$	$O(n^2 \log n)$	$\widetilde{\Theta}(\sqrt{p}) = \widetilde{\Theta}(2^{n/2})$	all $p > 3$
restricted one-shot (§5)	$\approx 2.5n$	$O(n^2 (\log n)^2)$	$p^{1/4+o(1)}$ SEA; $p^{1/2+o(1)}$ CM	all $p > 3$
one-shot, n^4-smooth [OneShot]	$\approx 2.5n$–$3n$	$O(n^2 (\log n)^2)$	$p^{1/8+o(1)}$ SEA; $p^{1/4+o(1)}$ CM	all $p > 3$
short ECPP [Short]	$O(n)$, $\lesssim 5n$	$O(n^2(\log n)^2)$; $O(n^2 \log n)^\dagger$	$L_p\big(1/3,\ (3/2)^{1/3} \approx 1.14\big)$, SEA & CM	all $p \geq 5$
ECPP [GK86, AM93]	$\Theta(n^2/\log n)$	$O(n^3)$	$(\log p)^{6+o(1)}$ SEA; $(\log p)^{4+\varepsilon}$ CM	all p (heur.)‡

TABLE 1. The certificate families. The Pomerance-triple verification is exactly $(\frac{5}{2} + o(1))\, n$ multiplications mod p. †The $O(n^2 \log n)$ bound holds for certificates whose fillers have polynomially many bits or few distinct prime factors — every published certificate, and every prime has one (prime-power fillers, §6.2); the unconditional worst case, attained by the $k = 0$ stratum (= the restricted one-shot), matches the one-shot row, and Remark 6.3 tightens the format to guarantee $O(n^2 \log n)$ outright. ‡Heuristic for all p; provable for almost all p. Primality is in ZPP via [AH92], but with certificates built on genus-2 Jacobians, not certificates of this format. The AKS–Bernstein row is discussed in Remark 1.1; the deterministic finding entry is the Lenstra–Pomerance decision bound, whose exponent c_0 is effectively computable but has never been made explicit. The Pratt entry is the cost of factoring $p - 1$ (and recursively below, a geometrically decreasing cost) by the number field sieve. The ECPP row's size and verification describe the customary smooth-cofactor descent ($\Theta(\log n)$ bits shed per step); the strict $m = 2q$ chain of [GK86] measures $\Theta(n^2)$ and $O(n^3 \log n)$ (§7.2).

Remark 1.1 (the AKS family, in certificate form). Agrawal, Kayal, and Saxena [AKS04] decide primality deterministically with no certificate at all, in time $(\log p)^{21/2+o(1)}$; Lenstra and Pomerance [LP19] reduce the exponent to 6: their Theorem 1 gives a deterministic algorithm running "in time at most $(\log n)^6 \cdot (2+\log\log n)^{c_0}$" for "some effectively computable real number c_0" — effectively computable in the sense that their proofs "implicitly contain an algorithm for computing c_0," which no one has carried out; so the sharpest citable deterministic bound is exactly $(\log p)^6(2+\log\log p)^{c_0}$, and the c in a formulation $O(n^6 \log^{c+o(1)} n)$ is not known. The *certificate* form of the family descends from Berrizbeitia [Berr05] through Cheng and Mihăilescu–Avanzi to Bernstein [Ber07] (see [Ber07] for the history and the precise definition): a certificate is a tuple (d, e, c, c_-, f, r, S) — f monic of degree d over $\mathbb{Z}/p\mathbb{Z}$, e a divisor of p^d-1, an element r and a finite witness set S in $R = (\mathbb{Z}/p\mathbb{Z})[y]/(f)$ — whose validity, checked by explicit congruences and unit tests in R, forces p to be a prime power by an AKS-style counting argument (a $(\log p)^{1+o(1)}$ perfect-power test [Ber98] finishes the job). Every prime has one with $S = \{1\}$, $d = (\log p)^{o(1)}$, and $e = (\log p)^{2+o(1)}$ [Ber07, Thms. 5.1–5.3], of total size $(\log p)^{1+o(1)}$ bits — i.e. $n^{1+o(1)}$, essentially the same order as the elliptic certificates — findable in *expected time* $(\log p)^{2+o(1)}$ (the only randomness is sampling an irreducible f and a witness r, both dense) and verifiable deterministically in $(\log p)^{4+o(1)}$ bit operations: the dominant check is the congruence $(x-s)^{p^d} = r^{(p^d-1)/e}\,x - s$ in $R[x]/(x^e - r)$ for each $s \in S$, one p^d-th powering in a ring of $(\log p)^{3+o(1)}$ bits. As a concrete instance, $(1, 57449, 28724, 16826, y, 2, \{1\})$ is a certificate for $2^{1024}+643$. Deterministically, trying candidates for f and r in a fixed order heuristically finds a certificate in the same $(\log p)^{2+o(1)}$ time (only the guarantee is lost); the proven deterministic route remains the [LP19] *decision* algorithm — it decides primality without producing a certificate — whence the two finding entries in the table. The find/verify asymmetry is inverted in this family — finding is far cheaper than verifying — and the inversion has a stark consequence: a verifier willing to use Las Vegas randomness gains *nothing* from the certificate, since given the empty string it can regenerate certificate data itself in expected $(\log p)^{2+o(1)}$ time and run the same $(\log p)^{4+o(1)}$ check (composite inputs are dispatched surely, and quickly, by strong-pseudoprime witnesses [Mil76, Rab80]). The certificate is information its receiver could just as cheaply have produced — its size is beside the point — and what it buys is *determinism* alone: a supplied tuple is checked with no randomness in $(\log p)^{4+o(1)}$, against the $(\log p)^6(2+\log\log p)^{c_0}$ certificate-free deterministic decision. The verification itself is stubbornly quasi-*quartic*: the soundness counting argument forces the $(\log p)^{3+o(1)}$-bit ring (its binomial condition needs $e\#S \gtrsim \sqrt{e}\, d \log p$, so $e = \widetilde{\Omega}(\log^2 p)$ for bounded $\#S$), no way to check the powering short of computing it is known, and larger $\#S$ trades a polylogarithmically smaller e against linearly more congruences, and loses — a floor for every known verifier, though not a proven lower bound. This family therefore does *not* subsume the quasi-quadratic verification of the elliptic formats, whose certificates transfer work the verifier could never afford; see the closing open problem of §8.

Two engines supply candidate curve orders throughout, and they trade off differently in every section: **SEA** point counting (sample a curve, compute its order; polynomial cost per candidate, orders statistically "random") and the **CM method** (enumerate imaginary quadratic discriminants D with $4p = t^2 + |D|v^2$; each discriminant is tested in polylogarithmic time, but discriminants up to B yield only $\widetilde{\Theta}(\sqrt{B})$ usable orders, and committing to a winner costs a class polynomial of degree $h(D) \approx \sqrt{|D|}$, computed in time $\widetilde{O}(|D|)$). Where the searched event is polynomially rare, CM's cheap candidates win; where it is exponentially rare, the $\sqrt{B}$ supply curve throttles CM and SEA's flat polynomial cost per candidate wins asymptotically. Measured crossovers appear in §4.3 and §5.2.

2. Preliminaries

2.1. **Cost model and notation.** p always denotes the integer whose primality is at issue (a "probable prime" during a search), and $n = \lceil \log_2 p \rceil$ its bit length. $M(t)$ denotes the bit cost of multiplying two t-bit integers; $M(t) = O(t \log t)$ unconditionally [HvdH21]. A "multiplication mod p" costs $O(M(n))$; following Pomerance we count squarings as multiplications and ignore additions. Every complexity bound below is stated in bit operations; multiplication counts appear only where an exact constant is the natural statement (as in Pomerance's theorem), convertible at 1 mult $= \Theta(M(n))$ bit operations. $\widetilde{O}(\cdot)$ suppresses polylogarithmic factors. The subexponential scale is

$$L_p(\alpha,\, c) \;=\; \exp\big((c + o(1))\,(\ln p)^{\alpha}(\ln \ln p)^{1-\alpha}\big),$$

so $L_p(0,\, c)$ is polynomial in $\log p$, $L_p(1,\, c)$ fully exponential, $L_p(1/2,\, \cdot)$ the ECM/quadratic-sieve scale, and $L_p(1/3,\, \cdot)$ the number field sieve scale. "$p^{\theta+o(1)}$" means $\exp((\theta + o(1)) \ln p)$; the $o(1)$ terms here go to zero slowly (they hide $\ln \ln$ factors in the exponent), which is why observed behavior at a few hundred bits is far friendlier than the exponent suggests — a point made quantitative in §4 and §5.

All *verification* statements below are rigorous, deterministic, and unconditional: a verifier that outputs True proves primality with no hypotheses. All *finding* statements are heuristic, resting on the two randomness assumptions of §2.4–2.5 (curve orders behave like uniform Hasse-interval integers up to explicit local corrections; smooth-part probabilities follow the Dickman distribution). This asymmetry — unconditional verification, heuristic search — is intrinsic to the subject and is the reason certificates matter at all: the expensive, unrigorous part need be run only once, by anyone, while the cheap, rigorous part can be repeated by everyone.

2.2. **Curves over $\mathbb{Z}/p\mathbb{Z}$ and Montgomery arithmetic.** For prime $p > 3$, an elliptic curve over $\mathbb{F}_p$ in Montgomery form is

$$E_{A,B}\colon\; By^2 = x^3 + Ax^2 + x, \qquad B(A^2 - 4) \neq 0,$$

with the usual chord–tangent group law on its projective points; $\#E$ denotes the number of points including the point at infinity O [Sil09, Mon87]. Every Montgomery curve has the rational 2-torsion point $(0, 0)$, so $4 \mid \#E$ always (the three 2-torsion x-coordinates are 0 and the roots of $x^2 + Ax + 1$; at least one lies on E or forces a point of order 4). Montgomery's x-only arithmetic works with classes $(X : Z)$ representing $x = X/Z$ on the *Kummer line*, identifying P with $-P$: doubling costs 5 multiplications (with $c = (A + 2)/4$ precomputed: $U = (X + Z)^2$, $V = (X - Z)^2$, $X' = UV$, $Z' = (U - V)(V + c(U - V))$), differential addition costs 6, and the Montgomery ladder computes $(X : Z)$ of $[k]P$ in $\approx 11 \log_2 k$ multiplications. Two features matter for certificates.

Twist-agnosticism. The x-only formulas depend only on A, not B, and are simultaneously valid on $E_{A,1}$ and its quadratic twist. Every $x_0 \in \mathbb{F}_p$ is the x-coordinate of a point on one of the two (which one depends on whether $x_0^3 + Ax_0^2 + x_0$ is a square). All four certificate formats therefore quantify over "there exist B, y_0 such that (x_0, y_0) lies on $By^2 = x^3 + Ax^2 + x$": the verifier never tests whether x_0 is "on the curve," because it is always on one of the pair, and the order condition is what carries the proof. This matches the set-up of [Pom87], whose homogeneous model $y^2z = x^3 + ax^2z + bxz^2$ is the general curve with a rational point of order 2.

Soundness over $\mathbb{Z}/p\mathbb{Z}$ for composite p. The verifier runs the same formulas modulo a p not yet known to be prime. The computation then commutes with reduction modulo every prime $\ell \mid p$ *provided no ring division by a zero-divisor occurs*; verifiers enforce this with gcd guards ($\gcd(A^2 - 4, p) = 1$; a claimed point at infinity must be $(X : 0)$ with $\gcd(X, p) = 1$; a claimed non-identity must have $\gcd(Z, p) = 1$). A failed guard either rejects or exhibits a proper factor of p. This is the same discipline as Lenstra's "pseudocurves" in ECM [Len87], run in reverse: ECM hopes for a zero-divisor, verification insists on their absence.

2.3. **The certification lemma.** All four families rest on one lemma, essentially Goldwasser–Kilian's [GK86], in the single-curve form of [Pom87]. Write

$$L(p) \;=\; q+1+\lfloor 2\sqrt{q}\rfloor, \qquad q=\lfloor \sqrt{p}\rfloor,$$

the integer form of $(p^{1/4}+1)^2$: by the Hasse bound [Sil09, V.1], $L(p)$ is an upper bound on the order of any point on any elliptic curve over any field $\mathbb{F}_\ell$ with $\ell \le \sqrt{p}$ (an upper bound is all the lemma below needs; since q need not be prime, the bound need not be attained).

Lemma 2.1 (elliptic certification). *Let $p>3$ be an odd integer, and suppose that for every prime ℓ dividing p, the reduction modulo ℓ of some fixed pair (A, x_0) yields a point of order exactly m on a Montgomery curve $By^2 = x^3 + Ax^2 + x$ over $\mathbb{F}_\ell$, for one and the same integer $m > L(p)$. Then p is prime.*

Proof. If p were composite it would have a prime divisor $\ell \le \sqrt{p}$. The point's order m divides $\#E(\mathbb{F}_\ell) \le \ell + 1 + 2\sqrt{\ell} \le L(p) < m$, a contradiction. □

Establishing "order exactly m modulo every prime divisor" from x-only data is the verifier's whole job: it checks that $[m](x_0{:}1)$ is the genuine point at infinity ($Z \equiv 0$ and $\gcd(X,p)=1$, which forces $[m]P = O$ modulo *every* $\ell \mid p$), and that $[m/q](x_0{:}1)$ is a unit-Z point (hence $\neq O$ modulo every $\ell \mid p$) for each maximal proper divisor m/q of m, q running over the distinct primes of m. This requires *knowing the prime factorization of m* — which is why every format constrains m's arithmetic shape, and why the differences between the formats are exactly differences in how m may be built.

Remark 2.2 (why "modulo every prime divisor" and not "order in $E(\mathbb{Z}/p\mathbb{Z})$"). The order of a point over the ring $\mathbb{Z}/p\mathbb{Z}$ is, by CRT, only the *least common multiple* of its orders modulo the primes dividing p; for composite p the required order can be split between factors with every local order strictly smaller than m. A concrete counterexample from [Short]: $p_0 = 4410667997551 = 2098153 \cdot 2102167$ with the tuple $(p_0, 1365834658413, 107710304518, 4200232, \ldots)$ has a point of order exactly $8 \cdot 525029$ in $E(\mathbb{Z}/p_0\mathbb{Z})$ — order 8 modulo the first prime and 525029 modulo the second — and would pass a verifier that read "order" in $\mathbb{Z}/p_0\mathbb{Z}$. The gcd guards are what rule this out: they force the ladder's intermediate values to behave identically modulo every prime divisor, so the order is m locally everywhere, and the reference verifiers reject the example (exposing both factors of p_0 along the way).

2.4. **The supply of curve orders.** Fix a prime p. Which integers occur as $\#E$ for $E/\mathbb{F}_p$, and how are they distributed? Three classical facts calibrate every heuristic in this write-up.

- **Hasse** [Sil09]. $\#E = p+1-t$ with $|t| \le 2\sqrt{p}$: orders live in the *Hasse interval* of length $4\sqrt{p}$ centered at $p+1$.
- **Deuring** [Deu41, Wat69]. Conversely *every* such t occurs for some $E/\mathbb{F}_p$ (p prime). Quantitatively, the number of curves with trace t, counted up to isomorphism and weighted naturally, is the Hurwitz class number $H(4p-t^2)$ [Deu41, Sch87], of typical size $\sqrt{p}$ up to logarithmic factors: a random curve has an essentially uniformly distributed trace, modulated by a smooth semicircular-type density and by class number fluctuations.
- **Group structure.** $E(\mathbb{F}_p) \cong \mathbb{Z}/n_1 \times \mathbb{Z}/n_2$ with $n_1 \mid n_2$ and $n_1 \mid p-1$; a point of order m exists iff $m \mid n_2$ (the exponent). For a curve chosen at random, n_1 is 1 or very small with high probability, and one can always move within the isogeny class to make any given Sylow subgroup cyclic (§2.6.2).

The randomness heuristic. For searching we model the order of a random curve as a uniform random integer in the Hasse interval, corrected by known local biases: $\#E$ is even with probability $\approx 2/3$ over all curves, and exactly 1 for Montgomery curves (which force $4 \mid \#E$); more generally

$\Pr[\ell \mid \#E] \approx \ell/(\ell^2-1)$ for small primes ℓ [Len87, Prop. 1.14]. These biases *help* every smoothness-hunting search below by a constant factor, and are one ingredient of the $O(1)$ "fudge factors" between the clean Dickman model and observed candidate counts (measured in §4.3 and §5.3 to lie between about 1/2 and 20).

2.5. **Smooth numbers.** An integer is *y-smooth* if all its prime factors are $\le y$. With $\Psi(x, y) = \#\{\, m \le x : m \text{ is } y\text{-smooth}\,\}$, the fundamental estimate [Dic30, CEP83, Hil86] is

$$\Psi(x,y) \;=\; x\,\rho(u)^{1+o(1)}, \qquad u = \ln x/\ln y,$$

uniformly for $u < y^{1-\varepsilon}$, where the Dickman function ρ satisfies $\rho(u) = 1$ on $[0, 1]$ and $u\rho'(u) = -\rho(u-1)$; asymptotically $\rho(u) = u^{-u(1+o(1))} = \exp(-u(\ln u + \ln\ln u - 1 + o(1)))$. See [Gra08] for a survey. Two consequences are used constantly.

Fully smooth orders. $\Pr$[a random Hasse-interval integer is y-smooth] $\approx \rho(u)$ with $u = \ln p/\ln y$. With y polylogarithmic in p, $u \approx \ln p/(c \ln\ln p)$ and $\rho(u) = p^{-1/c+o(1)}$: *polylog-smooth orders are a positive power of p rare.* This single computation drives all the exponents 1/2, 1/4, 1/8 below.

Smooth parts. Write $S_y(N)$ for the largest y-smooth divisor of N. The certificates of §§4–6 need only $S_y(\#E) > L(p) \approx \sqrt{p}$ — half the bits — not full smoothness. For $N \approx p$ random, $\Pr[S_y(N) > \sqrt{N}] = \rho(u/2)^{1+o(1)}$: up to factors absorbed in the $o(1)$ (an $e^{-\gamma}$ from the rough cofactor, an integral over how far above $\sqrt{N}$ the smooth part lands), demanding a smooth *part* above $\sqrt{N}$ costs what full smoothness would cost for $\sqrt{N}$. Halving the bits that must be smooth halves u, and since ρ is doubly exponential in u this is an enormous relaxation: at 256 bits with $y = n^4$ it is the difference between $\rho(8) \approx 3 \times 10^{-8}$ and $\rho(4) \approx 5 \times 10^{-3}$.

2.6. **Two engines for curve orders: SEA and CM.** Every finder below is a filter over candidate pairs (curve, order). Two general-purpose engines manufacture such pairs (special forms of p admit shortcuts, §4.3.4), and their cost structures are dual.

2.6.1. *Point counting: Schoof–Elkies–Atkin.* Schoof's algorithm [Sch85] computes $\#E$ for arbitrary $E/\mathbb{F}_p$ in deterministic polynomial time by computing the trace $t \bmod \ell$ for the primes $\ell = O(\log p)$, working in the ℓ-division polynomials of degree $\Theta(\ell^2)$; with fast arithmetic it runs in $\widetilde{O}(n^5)$. The Elkies–Atkin improvements [Sch95, Elk98] replace division polynomials, for the $\approx$half of primes ℓ that are "Elkies primes," by degree-$\Theta(\ell)$ kernel polynomials of rational ℓ-isogenies located via the modular polynomial $\Phi_\ell(X, j(E))$, giving the SEA algorithm with heuristic cost $\widetilde{O}(n^4)$ (the heuristic being the supply of Elkies primes; see [SS14] for what is provable on average). As an order *supply*, SEA has two decisive properties: the cost per candidate is a *flat polynomial* — independent of how many candidates one consumes — and each quadratic twist comes free ($\#E + \#E^{\text{tw}} = 2p + 2$), so one count yields two candidate orders. Its constant is large in practice: a stock PARI/GP `ellcard` costs 7–14 core-seconds per curve at 384–448 bits [OSF].

2.6.2. *Complex multiplication: prescribed traces from small discriminants.* The CM method inverts the problem: instead of choosing a curve and learning its order, choose the order and construct a curve. If $D < 0$ is an imaginary quadratic discriminant with

$$4p = t^2 + |D|v^2$$

solvable in integers (equivalently, p splits suitably in the ring class field of discriminant D), then there are curves $E/\mathbb{F}_p$ with CM by the order of discriminant D and trace $\pm t$, hence orders $p+1-t$ and $p+1+t$; for $D = -3, -4$ the extra units give six resp. four orders [AM93, Cox13]. The machinery has three separately priced stages.

(1) *Which D are usable, and what are t, v?* Testing one D costs a Legendre symbol, a square root of $D \bmod p$, and Cornacchia's algorithm [Cor08], [CP05, §2.3.4] — polylogarithmic. Batched over many D, the square roots reduce to one square root per prime factor

of the discriminants plus one modular multiplication per D (Shallit's trick; credited in [Mor07, FKMW04]), and restricting to discriminants all of whose prime discriminant factors are residues on p boosts the solvability probability by genus theory ($\Pr = 2^{\#\text{factors}-1}/h(D)$ for such D). The engine of [OSF] scans 5–20 million discriminants per second at 128–384 bits this way.

(2) *How many orders does a budget of discriminants buy?* This is the binding constraint. The solvability probability for a single D is $\approx c/h(D) \approx c'/\sqrt{|D|}$ (up to $L(1,\chi_D)$ factors), so discriminants up to B yield only

$$\#\{\text{usable } (D,t)\} \;\approx\; \sum_{|D|\le B} \Theta(1/\sqrt{|D|}) \;\approx\; \widetilde{\Theta}(\sqrt{B})$$

candidate orders, at scan cost $\widetilde{\Theta}(B)$. Empirically the constant is ≈ 0.5: the run of [OSF] on $10^{110}+7$ measured a pool of 39,878 orders at $B = 8\times 10^9$, i.e. $\approx 0.45\sqrt{B}$. *Producing K candidate orders by CM costs $\widetilde{\Theta}(K^2)$.*

(3) *Committing to a winner.* Only the finally selected D requires the expensive step: the Hilbert class polynomial H_D (or a smaller class invariant) of degree $h(D) \approx \sqrt{|D|}$, computable in quasi-linear time $\widetilde{O}(|D|)$ by the CRT method [Sut11, Eng09, Sut12], one root of it mod p, the j-invariant, the right twist, and a point of the desired order. Isogeny-volcano navigation [Koh96, FM02, Sut13] fixes group-structure obstructions (§4.3).

The duality, which recurs in every section: **SEA pays** $\mathrm{poly}(n)$ **per candidate forever; CM pays polylog per discriminant tested but quadratically in the number of candidate orders delivered.** If a search needs K candidates, SEA costs $\widetilde{O}(K)\,\mathrm{poly}(n)$ and CM costs $\widetilde{O}(K^2) + \widetilde{O}(K\,\mathrm{polylog})$ — the K^2 paid twice over, by the scan and again by the winner's class polynomial ($|D| \approx K^2$). CM wins when K is modest (all of classical ECPP, one-shot to ≈ 450 bits); SEA wins when K explodes (one-shot beyond ≈ 500 bits, the restricted variant, and — in the degenerate single-candidate sense of §3 — never for Pomerance triples, where the admissible orders are known in advance and neither engine is needed).

2.7. **The prover's factoring toolbox.** Several finders must partially factor candidate orders (never as part of any certificate's validity — only to *recognize* a usable order). Trial division to y costs $\pi(y)$ divisions per candidate naively, and amortizes to polylogarithmic per candidate over large batches via product/remainder trees [Ber04]; ECM [Len87] finds a prime factor $q \le y$ of N in heuristic expected time $L_y(1/2,\sqrt{2})\cdot \widetilde{O}(M(\log N))$, so "peel everything $\le y$ from N" costs $L_y(1/2,\sqrt{2})$ up to polynomial factors; the number field sieve [LL93] factors N completely in heuristic $L_N(1/3,(64/9)^{1/3} \approx 1.923)$. The interplay "how much peeling per candidate vs. how rare is a peelable candidate" is what produces the $L(1/3)$ finding costs of §6.

3. Pomerance proofs

3.1. **The problem.** Pomerance's 1987 theorem [Pom87, Thm. 1]: *for every prime p there is a proof of its primality whose verification requires* $(5/2+o(1))\log_2 p$ *multiplications* mod p. His certificate exhibits a point of order 2^k on an elliptic curve, with 2^k just above the threshold $L(p)$ of §2.3, so that verification is nothing but $k \approx \frac{1}{2}\log_2 p$ successive doublings. The DANGER3 repository [DANGER3] (the data challenge of the April 2026 BIRS workshop *DANGER: Data, Numbers, and Geometry*) fixes the modern Montgomery-form refinement:

Definition 3.1 ([DANGER3])**.** A *Pomerance triple* is a triple of integers (p, A, x_0) with p a positive odd integer, $0 \le A, x_0 < p$, $A \not\equiv \pm 2 \bmod p$, such that there exist B and y_0 making (x_0, y_0) a point of order 2^k on the Montgomery curve $By^2 = x^3 + Ax^2 + x$ modulo every prime divisor of p, where k is the *least* integer with $2^k > q + 1 + 2\sqrt{q}$, $q = \lfloor\sqrt{p}\rfloor$. Operationally: applying the

Montgomery doubling law $k-1$ times to $(x_0 : 1)$ gives a point with Z-coordinate coprime to p, and the k-th doubling gives $Z \equiv 0 \pmod{p}$.

By Lemma 2.1 a valid triple proves p prime ($2^k > L(p)$), and by [Pom87, Lemmas 2.1–2.2] the doubling-chain behavior ($Z \neq 0$ at step $k-1$, $Z = 0$ at step k) characterizes "order exactly 2^k" on the Kummer line. The certificate is *two* integers beyond p itself — about $2n$ bits — the shortest of the elliptic-curve formats.

3.2. **Existence: every prime has one.** This is the part that uses the full strength of Deuring's theorem, and it is worth sketching because every other format inherits its existence statement from this argument [Pom87, Thm. 3.3].

(1) Minimality puts 2^k in $(L(p), 2L(p)]$, which is below the Hasse interval's width $4\sqrt{p}$ once $p > 34$, so the interval $(p+1-2\sqrt{p},\, p+1+2\sqrt{p})$ contains a multiple $m = 2^k c$ of 2^k.
(2) By Deuring [Deu41], some $E/\mathbb{F}_p$ has $\#E = m$ (m is even, so E can be taken with a rational 2-torsion point, i.e. in the model of §2.2).
(3) $E(\mathbb{F}_p)$ is a product of at most two cyclic groups. If some curve of order m has a point of order 2^k, done (Pomerance's "type 1" certificate). Otherwise the 2-Sylow subgroup splits; Pomerance's "type 2" certificate exhibits two points of orders $2^{k_1}, 2^{k_2}$ with $k_1 + k_2 = k$ generating a subgroup of order 2^k, which suffices for the same contradiction. For the single-point Montgomery format one instead moves within the isogeny class: descending the 2-isogeny volcano to its floor [Koh96, Sut13] makes the 2-Sylow subgroup cyclic, so a point of order 2^k exists there, and the floor curve is Montgomery-representable. This is the "minor refinement" recorded in [DANGER3]: *Pomerance triples exist for all primes* $p > 3$.

Empirically the repository verifies this exhaustively far beyond the theorem's needs: all 43 million triples with $p < 2^{14}$, and at least one triple for *every* prime $3 < p < 2^{32}$ (203,280,219 of them).

3.3. **Verification, exactly.** The reference verifier `vpp.py` [DANGER3] computes $k =$ bit-length of $q + 1 + \lfloor 2\sqrt{q} \rfloor$, checks $\gcd(A^2 - 4, p) = 1$, and runs k Montgomery doublings from $(x_0 : 1)$ at 5 multiplications each (§2.2), accepting iff $Z_k \equiv 0$ and $\gcd(Z_{k-1}, p) = 1$. Total:

$$5k + O(1) \;=\; (5/2 + o(1)) \log_2 p \;\text{ multiplications mod } p,$$

i.e. $O(n\,M(n)) = O(n^2 \log n)$ bit operations, plus two gcds. This realizes Pomerance's constant: his Theorem 3.3 proves $(7/2)\log_2 p + O(1)$ multiplications with the 7-multiplication doubling of his general two-parameter model, and the remarks following it reduce $7/2$ to $5/2$ — precisely the 5-multiplication Montgomery doubling (Montgomery's paper [Mon87] appears in the same volume of *Mathematics of Computation*). For calibration: a *single* strong-pseudoprime test — which proves nothing — costs $(1 + o(1))\log_2 p$ multiplications; a Pomerance triple proves primality outright for $2.5\times$ that. The repository also provides `lean_vpp.py`, which emits a Lean 4 script certifying a given triple, so verification can even be delegated to a proof assistant's kernel.

3.4. **Finding certificates: $\widetilde{\Theta}(\sqrt{p})$, and why.** Here the supply engines of §2.6 are both nearly useless, which is what makes this family the hardest to find and the sharpest illustration of the find/verify gap.

The counting heuristic. A curve is usable iff $\#E$ is one of the multiples of 2^k in the Hasse interval — two to four integers, *computable in advance from p alone* since $2^k \in (L(p), 2L(p)] \approx (1 \ldots 2)\sqrt{p}$ — and iff its group has a cyclic-enough 2-Sylow subgroup with the sampled point generating its full 2-part. Under §2.4 each admissible order is hit with probability $\widetilde{\Theta}(1/\sqrt{p})$, and the structure conditions cost only a constant, so the per-curve success probability is $\widetilde{\Theta}(1/\sqrt{p})$, the hidden constant wobbling by a factor ≈ 4 with the position of $\sqrt{p}$ between consecutive powers of two.

So $\widetilde{\Theta}(\sqrt{p})$ curves must be sampled, and the algorithmic content is making the per-candidate test essentially free.

The 2-Sylow projection test (Ruehle's implementation [Rue26], used for the 10^{19} and 10^{21} records). Let $N_1, \ldots, N_j$ ($j \leq 4$) be the admissible orders and $c_i = N_i/2^{v_2(N_i)}$ their odd parts. For a random pair (A, x_0): one Montgomery ladder computes $[\prod c_i](x_0 : 1)$, which lands in the 2-Sylow subgroup *whenever the curve's order is admissible*; then repeated doublings with early abort either die (reject) or reach $Z = 0$ at exactly depth k (accept, and the affine x-coordinate of the point k steps back is the certificate's x_0). Without the projection, a random point has 2-power order with probability only $\widetilde{\Theta}(1/p)$ — the projection is what turns p trials into $\sqrt{p}$ trials. Per-candidate cost: one ladder of $O(n)$ bits plus $O(k)$ cheap doublings, i.e. $O(n)$ multiplications on numbers that fit in one or two machine words at challenge sizes — a few hundred nanoseconds on a CPU core, a few nanoseconds amortized on a GPU. Sampling curve families with prescribed 2-power torsion (via the modular curve $X_1(16)$) conditions $v_2(\#E)$ and buys a further constant factor, as observed in the 10^{23} search [McL26] — explicitly a constant-factor gain, not an asymptotic one.

Why neither engine helps. SEA point counting per candidate ($\widetilde{O}(n^4)$ each) is strictly worse than the $\widetilde{O}(n)$-multiplication direct test, and no partial-information shortcut is useful: $t \bmod 2^j$ for j up to $k \approx n/2$ is exactly as hard as the event being tested. The CM route would need a small discriminant D with $4p = t^2 + |D|v^2$ and $t \equiv p + 1 \pmod{2^k}$; only $O(1)$ values of t satisfy the congruence inside the Hasse range, and for fixed t the probability that $4p - t^2$ has squarefull part large enough to leave $|D| \leq X$ is $\approx \sqrt{X/p}$ — total expected work $p^{1/2-o(1)}$ for every parameter choice (a heuristic accounting of the natural CM enumerations, not an algorithm-independent lower bound). Isogeny walks preserve the order, torsion families fix only $O(1)$ bits of $v_2(\#E)$, and the admissible set has no multiplicative structure to sieve on. Nothing better than $\widetilde{\Theta}(\sqrt{p})$ is known, and the DANGER3 progression (below) is the empirical record of that $\sqrt{p}$ wall being pushed one decimal digit — half a bit of $\sqrt{p}$ — at a time.

3.5. **Results to date.** [DANGER3] lists example triples for $p = 10^{12} + 39$ through $10^{18} + 3$, then a challenge log in which every entry is a human–LLM collaboration (Table 2). Candidates are (A, x_0) pairs tested by the projection method; $\sqrt{p}$ is shown for comparison with the $\widetilde{\Theta}(\sqrt{p})$ heuristic.

p	$\sqrt{p}$	candidates	time / hardware	found by (date)
$10^{19} + 51$	3.2×10^9	3.7×10^9	400 s, multithreaded C	F. Ruehle + Claude Opus 4.6 (Apr 2026)
$10^{20} + 39$	1.0×10^{10}	1.2×10^9	7 h	V. Jejjala + GPT 5.4 Pro (Apr 2026)
$10^{21} + 117$	3.2×10^{10}	5.3×10^{10}	16 h	F. Ruehle + Claude Opus 4.6 (Apr 2026)
$10^{22} + 9$	1.0×10^{11}	5.9×10^{10}	16 h	A. McLain + GPT 5.5 Codex (May 2026)
$10^{23} + 117$	3.2×10^{11}	3.1×10^{10}	—	A. McLain + GPT 5.5 Codex (Jun 2026)
$10^{24} + 7$	1.0×10^{12}	1.8×10^{10}	6 h, Apple M3 laptop	J. Shi + Claude Fable 5 (Jun 2026)
$10^{25} + 13$	3.2×10^{12}	2.0×10^{11}	—	A. McLain + GPT 5.5 Codex (Jun 2026)
$10^{26} + 67$	1.0×10^{13}	1.4×10^{11}	45 min, RTX 6000 GPU	A. McLain + GPT 5.5 Codex (Jun 2026)

TABLE 2. The DANGER3 challenge log for Pomerance triples.

The observed candidate counts sit between $0.01\sqrt{p}$ and $1.7\sqrt{p}$ — the spread reflecting the $2^k/\sqrt{p}$ wobble, torsion conditioning, twist usage, and single-draw geometric variance — and the per-candidate cost has been driven from CPU-microseconds to GPU-nanoseconds. The open challenge as of this writing is $p = 10^{27} + 103$: $\sqrt{p} \approx 3.2 \times 10^{13}$ mean-case candidates, roughly 170 RTX-6000-hours at the measured 10^{26} rate of 5×10^7 candidates per second — a GPU-week, though the log's variance (the 10^{26} entry won at $0.014\sqrt{p}$) leaves anything from hours on up: within reach. Note the striking asymmetry these records quantify: a certificate that takes GPU-*hours* to find is verified in ≈ 220 multiplications of 90-bit numbers — microseconds.

4. One-shot ECPP

4.1. **The problem.** The OneShotPrimalityProofs repository [OneShot] (part of the DARPA expMath program) generalizes the Pomerance triple by relaxing "order a power of two" to "order n^4-smooth," keeping the single curve — one shot, no recursion — and the quasi-quadratic verification.

Definition 4.1 ([OneShot])**.** A *one-shot ECPP* is a tuple $(p, A, x_0, m, q_1, \ldots, q_k)$ with $p > 1$ odd, $0 \le A, x_0 < p$, $A \not\equiv \pm 2 \bmod p$, such that (x_0, y_0) is a point of order m on $By^2 = x^3 + Ax^2 + x$ (for some B, y_0), where, with $n = \lceil \log_2 p \rceil$ and $L = L(p)$: m is n^4*-smooth*; $L < m < L \cdot r$ where r is the least prime divisor of m; and $q_1 < \cdots < q_k$ are the prime divisors of m in (n^2, n^4).

Every Pomerance triple is the case $k = 0$, $m = 2^k$ minimal — so one-shot certificates exist for every prime $p > 3$, unconditionally. The minimality window $m < L \cdot r$ generalizes "least power of two above L": it forces $m/r \le L$, i.e. removing the smallest prime of m would fall out of the window, keeping $\log m = \frac{1}{2} \log p + O(\log n)$ and the certificate canonical-small. Since each listed prime exceeds n^2, $k < n/(4 \log_2 n) + O(1)$; and since the q_i divide m, their combined length is at most $\log_2 m \approx n/2$: the whole certificate is $\approx 2.5n$–$3n$ bits.

4.2. **Verification.** The reference verifier `voneshot.py` [OneShot] establishes "order exactly m modulo every prime divisor of p" and the window conditions, as follows.

(1) *Window:* $L < m \le p + 1 + \lfloor 2\sqrt{p} \rfloor$ (a point's order cannot exceed the Hasse bound if p is prime; soundness rests on Lemma 2.1, not on this check alone).
(2) *Nonsingularity:* $\gcd(A^2 - 4, p) = 1$.
(3) *Factor m:* sieve the primes to n^2 and extract m's prime divisors below n^2 by batched remainder trees [Ber04]; strip them to get the n^2-rough part m'; then check that the listed q_i are strictly increasing, lie in (n^2, n^4), and *exactly exhaust* m'. Two design points deserve emphasis:
   - *The q_i need no primality tests.* Any divisor of m' below n^4 is automatically prime: a composite one would have two factors $> n^2$, exceeding n^4. The certificate format chose the exponent 4 precisely to make the listed primes self-certifying.
   - *The q_i are validated against m', not m.* Validating against m would admit a "fold attack": a repeated small prime $c \le n^2$ of m could be folded into a fake "large prime" cq dividing m but not m', corrupting the exact-order argument. (A concrete attack tuple rejected by this check is in the verifier's test suite.)
(4) *Minimality:* $m < L \cdot r$ with r the least prime found.
(5) $[m]P = O$ *genuinely:* the ladder value $(X : Z)$ at m must have $Z \equiv 0$ *and* $\gcd(X, p) = 1$ — the second condition rejects the degenerate $(0 : 0)$ that arises if the ladder ran past the local order modulo some prime divisor of p, and is what makes the step sound over $\mathbb{Z}/p\mathbb{Z}$ (§2.3).
(6) *Exact order:* with $R = \prod q$ over the distinct primes $q \mid m$, compute $Q = [m/R]P$ once, then verify $[m/q]P \ne O$ for every q by a balanced divide-and-conquer over the prime set: split the primes in halves, multiply Q by each half's product, recurse on the opposite half. Each of the $O(\log n)$ recursion levels costs $O(n)$ group operations in total.

Proposition 4.2 (verification cost, rigorous)**.** *$O(n \log n)$ group operations, each $O(1)$ multiplications* mod p, *plus $\widetilde{O}(n^2)$ bit operations of sieving and trial division: $O(n^2(\log n)^2)$ bit operations and $O(n^2)$ bits of memory — quasi-quadratic, a factor $\approx \log\log p$ above Pomerance triples (the depth of the product tree), against quasi-cubic for a classical ECPP chain (§7). Measured* [OSF]*:*

8–14 ms in pure Python at 256–333 bits. This is the worst case over certificates; §6.2 gives the accounting parametrized by the number of distinct primes of m, which recovers $O(n^2 \log n)$ when that number is bounded.

4.3. **Finding certificates.** Since m divides $\#E$ and every divisor of the n^4-smooth part of $\#E$ is reachable as a point order somewhere in the isogeny class (the group-structure corrections handled below), a candidate order is usable precisely when

$$S_{n^4}(\#E) \;>\; L \;\approx\; \sqrt{p},$$

in which case the minimal in-window divisor m of the smooth part is extracted greedily (strip the largest primes while staying above L; the greedy endpoint automatically satisfies $m < L \cdot r$). By §2.5 the per-candidate probability is $\rho(u/2)^{1+o(1)}$ with $u = \ln p / \ln n^4 = \ln p/(4 \ln\ln p)(1 + o(1))$, and

$$\rho(u/2) = \exp\bigl(-(u/2)(\ln u)(1 + o(1))\bigr) = p^{-1/8+o(1)},$$

so $p^{1/8+o(1)}$ *candidate orders* are needed. The $o(1)$ is strongly negative at practical sizes: $1/\rho(u/2)$ is only ≈ 200 at 256 bits, 1.6×10^4 at 384, 1.6×10^5 at 448, $\approx 3 \times 10^8$ at 640 bits. What the two engines pay per candidate then separates them.

4.3.1. *SEA sampling: $p^{1/8+o(1)}$ end-to-end.* The repository's `oneshot.gp` and the race harness `sea_supply.py` [OSF] sample random Montgomery coefficients A, compute $\#E$ by SEA (both twists per count), batch-test the smoothness gate, and assemble the winner. Per candidate: half an SEA count ($\widetilde{O}(n^4)$ heuristic) plus polylogarithmic gate work, so the total is $p^{1/8+o(1)} \cdot \mathrm{poly}(n) = p^{1/8+o(1)}$. The measured per-candidate cost with stock PARI (`ellcard`) is flat: 7.2 / 10.1 / 13.7 core-seconds at 384 / 416 / 448 bits.

4.3.2. *The CM engine: $p^{1/4+o(1)}$ end-to-end, far cheaper per candidate today.* The OneShot-FastECPP engine [OSF] is a fastECPP-style pipeline specialized to the one-shot problem — unusual in that its p is small (hundreds of bits) while its discriminants are huge ($|D|$ up to 10^{11}), the opposite of classical fastECPP, which relocates all the costs:

(1) *Discriminant scan*: enumerate every fundamental D with $|D| < B$ all of whose prime discriminant factors are residues on p (genus theory maximizes the solvability rate and keeps $\sqrt{D} \bmod p$ computable multiplicatively from a factor base of $\sqrt{q^*}$ values — one Tonelli–Shanks per factor-base prime, one modular multiplication per D, Shallit-style); run the fast-path Cornacchia on each. 5–20 million discriminants per second on 16 cores; yield $\widetilde{\Theta}(\sqrt{B})$ orders for cost $\widetilde{\Theta}(B)$.

(2) *Batched smoothness*: maintain each candidate's running n^4-smooth part across an adaptive ladder of cached prime-product segments starting just above n^2 (Bernstein remainder trees [Ber04]; the ladder deepens toward n^4 or widens the discriminant pool by a work-balance trigger, and winners usually appear octaves below n^4).

(3) *Commit*: for the winner (D, t), compute the class polynomial for the best class invariant (25–50× smaller than the Hilbert polynomial [Sut12]) by the CRT method [Sut11], parallelized across CRT primes; find one root mod p; map invariant $\to j$; assemble the Montgomery curve and a point of exact order m. Group-structure obstructions (a Montgomery model needs $4 \mid N$, and additionally $8 \mid N$ when $p \equiv 3 \bmod 4$; a point of order m needs $m \mid$ exponent) are *removed rather than filtered*: for each prime ℓ with non-cyclic ℓ-torsion, walk the ℓ-isogeny volcano to its floor [Sut13, FM02], where the ℓ-Sylow subgroup is cyclic and the curve is always representable.

Because the scan must reach $B \approx (\#\text{candidates})^2$, the end-to-end heuristic cost is $p^{1/4+o(1)}$ even though only $p^{1/8+o(1)}$ candidates are consumed — and the winner's commit stage costs the same: the winning discriminant is heuristically a uniform draw from the pool, so $|D| = \Theta(B) = p^{1/4+o(1)}$

typically, and its class polynomial costs $\widetilde{O}(|D|) = p^{1/4+o(1)}$ (§2.6.2). Only the certificate-facing objects live at the square-root scale: the degree $h(D) \approx \sqrt{|D|} \approx p^{1/8}$, the polynomial's $h \cdot n$ bits modulo p, and the root-finding, $\widetilde{O}(h)$ multiplications mod p (the largest committed class polynomial to date had $h = 35{,}085$ at 10^{100}). Direct enumeration of solutions $(t, v) \to D$ is worse ($\widetilde{\Theta}(p^{3/8})$ over the v-range that matters); closing the gap to $p^{1/8}$ would require *both* enumerating usable (D, t, v) in time proportional to their number *and* committing without a quasi-linear-in-$|D|$ class polynomial computation — neither is currently known, and the pair is the natural open problem of this section (§8, item 3).

4.3.3. *The measured SEA/CM crossover.* A head-to-head race on the same primes [OSF], 192-core machines, everything verified by `voneshot.py` (Table 3).

bits	$u/2$	$1/\rho(u/2)$	CM wall	SEA wall	CM cands (pool)	SEA cands	SEA vs CM (core-s/cand)
256	4.00	2.0×10^2	4.1 s	68 s	6,644	728	— (16-core box)
384	5.59	1.6×10^4	2,306 s	615 s	126,282	16,000	7.2 vs 3.5
416	5.97	4.8×10^4	1,774 s	2,805 s	152,484	52,000	10.1 vs 2.2
448	6.36	1.6×10^5	7,182 s	(stopped, 20,035 s)	309,264	$\geq 275{,}600$	13.7 vs 4.5

TABLE 3. The SEA-vs-CM one-shot race [OSF].

Individual walls flip on winner-density luck (the density draw has exponential-scale variance), but the per-candidate costs are unambiguous: SEA pays a flat 7–14 core-seconds while CM's average — mostly-linear segment building plus the quadratically growing scan and winner class polynomial — was still 2–4.5 core-seconds at pools up to 3×10^5. Extrapolating both curves puts the plain-SEA crossover at roughly **480–512 bits**; SEA-side accelerations that need SEA internals (an early-abort gate from the order modulo small ℓ during the Elkies phase, batch tails, torsion-family bias; modeled $\times$3–5) would pull it toward ≈ 380 bits, while the CM route's scan and winner-classpoly economics cap it near ≈ 500 bits regardless of patience. Beyond the crossover SEA sampling continues as $p^{1/8+o(1)}$ until the density itself walls out around 640 bits ($\approx 3 \times 10^8$ candidates $\times \approx 20$ core-seconds $\approx$ two hundred core-years). The observed winner densities track the Dickman model within a factor of 1–8 across all measured sizes — the "fudge" collecting the window/multiplicity/torsion structure conditions and sampling variance.

4.3.4. *Special forms: algebraic shortcuts.* Two independent contributions to the challenge table exploit structure instead of search [OneShot]:

- *The supersingular shortcut* (McLain + Claude, NIST P-384 and P-521). If $p \equiv 3 \bmod 4$, the curve $y^2 = x^3 + x$ is supersingular with order exactly $p + 1$ and is already Montgomery ($A = 0$); a certificate exists iff $S_{n^4}(p + 1) > L$, with no discriminant search and no class polynomial. Generalized-Mersenne primes make this likely because $p + 1$ factors algebraically: for P-384, $p + 1 = 2^{32}(2^{64} - 1)(2^{288} + 2^{224} + 2^{160} + 2^{96} - 1)$ has a 197-bit smooth part against a 193-bit L. For $p = 2^{521} - 1$ (P-521), $p + 1 = 2^{521}$ is entirely smooth and the certificate is a power-of-two Pomerance triple with $m = 2^{261}$. (For $p \equiv 1 \bmod 4$ the route is impossible in this format: $p + 1 \equiv 2 \bmod 4$ while every Montgomery order is divisible by 4.) This is the elliptic analogue of the classical $p + 1$ tests of [BLS75].
- *Reverse-CM order generation* (McLain + Claude Fable 5, $10^{80} + 129$, 25 CPU-minutes): the same architecture as §4.3.2 rediscovered independently at small scale — `qfbcornacchia` over $D \leq 8 \times 10^6$, an $8 \mid N$ intake filter for Montgomery representability, budgeted Brent–rho smooth parts, a subset-product search for m with exponents capped by $v_q(m) \leq v_q(N) - \min(\lfloor v_q(N)/2 \rfloor, v_q(p - 1))$ so that m divides the group exponent for every possible group structure, and one `polclass` for the winner.

4.4. **Results to date.** The challenge table [OneShot] holds verified certificates for the least prime above 10^c for $c = 27, \ldots, 120$ and for cryptographically famous primes. Selected entries, all verified by `voneshot.py`, appear in Table 4.

p	bits	route	time / resources
$10^{27} + 103 \ldots 10^{50} + 151$	90–167	SEA sampling (`oneshot.gp`)	2–170 CPU-s each
$10^{55} + 21 \ldots 10^{75} + 129$	183–250	CM engine [OSF]	≈1–7 s on 16 cores
$2^{255} - 19$ (Curve25519)	255	CM engine	≈4 s on 16 cores
P-256; secp256k1 field primes	256	CM engine	≈11 s; ≈8 s (16 cores)
$10^{80} + 129$	266	reverse-CM (independent)	≈25 CPU-min
$10^{85} + 103 \ldots 10^{105} + 3$	283–349	CM engine	24–500 s on 16 cores
$10^{110} + 7$, $10^{115} + 79$, $10^{120} + 79$	366–399	CM engine	121–226 s on 192–256 cores
BLS12-381 field prime	381	CM engine	≈18 s on 256 cores
NIST P-384 field prime	384	supersingular shortcut	<1 s, no search
$2^{521} - 1$ (P-521)	521	supersingular; $m = 2^{261}$	<1 s, no search

TABLE 4. Selected one-shot certificates from the challenge table [OneShot].

The frontier for *generic* primes stands just below 400 bits ($10^{120} + 79$, 2 minutes on 256 cores); the race of §4.3.3 reached 448 bits in two spot-hours, and the density model prices generic 512–640-bit certificates at tens to hundreds of core-years — possible, not yet done. Special-form primes (smooth $p + 1$) escape the wall entirely, as P-521 shows at 521 bits.

5. Restricted one-shot ECPP: m must be n^2-smooth

5.1. **The problem, and where it sits in the hierarchy.** Now forbid the listed primes: the certificate is (p, A, x_0, m) with *every* prime factor of m below n^2 — the $k = 0$ stratum of §4, equivalently a zero-level short ECPP (§6). The verifier is unchanged (`voneshot.py` with an empty list): it factors m completely by trial division below n^2 and needs no supplied data beyond m itself. Existence for every $p > 3$ again follows from Pomerance triples (2 is n^2-smooth); the certificate is $\approx 2.5n$ bits. Verification costs $O((n/2)(1 + \log k)\, M(n) + M(n^2))$ bit operations for a certificate whose m has k distinct primes (§6.2): Pomerance-order $O(n^2 \log n)$ for bounded k — a Pomerance triple is the case $k = 1$ — and $O(n^2 (\log n)^2)$ in the worst case $k = \Theta(n/\log n)$. What changes — dramatically — is the search.

5.2. **Finding: the exponent doubles.** The usable-curve condition becomes $S_{n^2}(\#E) > L$: the same half of $\#E$'s bits must now be built from primes below n^2 instead of n^4. The Dickman parameter doubles ($u = \ln p / \ln n^2 = 2 \ln p / (4 \ln \ln p)(1 + o(1))$), so the per-candidate probability is

$$\rho(u/2)^{1+o(1)} \;=\; p^{-1/4+o(1)},$$

the square of the n^4 density: $p^{1/4+o(1)}$ **candidates via SEA sampling, and** $p^{1/2+o(1)}$ **end-to-end for the CM route** (scan and winner class polynomial alike, at $B \approx \text{candidates}^2$) — the CM engine loses its practical edge entirely here, degenerating to the same $\sqrt{p}$ order as the Pomerance brute force, while SEA sampling stays genuinely sub-$\sqrt{p}$. Concretely, with the race's measured SEA rates (Table 5):

The n^4 window with its self-certifying listed primes is thus worth a full halving of the exponent ($1/4 \to 1/8$) at zero asymptotic cost to the verifier — the cleanest illustration in the family of how a small amount of certificate data (the q_i, a few hundred bits) buys an enormous amount of findability.

bits	$u/2$ ($y{=}n^2$)	$1/\rho(u/2)$	SEA cost estimate
67 (10^{20})	2.74	12	seconds, one core
100 (10^{30})	3.75	1.1×10^2	seconds, one core
133 (10^{40})	4.71	1.3×10^3	minutes, one core
167 (10^{50})	5.62	1.7×10^4	≈1 h, one core
200 (10^{60})	6.52	2.5×10^5	≈100 core-h
256	8.00	3.1×10^7	≈5–10 core-years (fleet-scale)
320	9.61	$\approx 9 \times 10^9$	$\approx 10^3$–10^4 core-years
384	11.18	$\approx 3 \times 10^{12}$	out of reach

TABLE 5. The feasibility frontier of the restricted variant.

5.3. **Results: first generic certificates.** The only previously existing restricted certificate at challenge scale was the special-form P-521 entry ($m = 2^{261}$, from $p + 1 = 2^{521}$). For this write-up we produced generic restricted certificates for the least primes above $10^{20}, \ldots, 10^{50}$ by SEA sampling (a 15-line wrapper over the repository's `oneshot.gp` with the smoothness bound set to n^2; first-winner-wins parallel workers; every certificate verified by `voneshot.py` with an empty prime list). Observed candidate counts sit within an order of magnitude of $1/\rho(u/2)$, as expected given the $O(1)$ structure corrections and geometric variance (Table 6).

p	bits	predicted	curves	time / cores	m of the certificate
$10^{20} + 39$	67	12	213	33 s, 1 core	$2^3{\cdot}5^3{\cdot}7{\cdot}13{\cdot}37{\cdot}47{\cdot}89$
$10^{30} + 57$	100	1.1×10^2	117	33 s, 1 core	$2{\cdot}7^3{\cdot}43{\cdot}109{\cdot}241{\cdot}251{\cdot}6899$
$10^{40} + 121$	133	1.3×10^3	$\approx 8 \times 10^3$	17 min, 8 workers	$2{\cdot}17{\cdot}37{\cdot}73{\cdot}197{\cdot}283{\cdot}673{\cdot}2797{\cdot}11827$
$10^{50} + 151$	167	1.7×10^4	$\approx 5.3 \times 10^4$	≈2 core-h	$2{\cdot}3{\cdot}7{\cdot}29{\cdot}37{\cdot}211{\cdot}337{\cdot}347{\cdot}449{\cdot}467{\cdot}3527{\cdot}12613$
$10^{60} + 7$	200	2.5×10^5	not run to completion (4.8×10^3 curves, no winner); $\approx 10^2$–10^3 core-h predicted		

TABLE 6. First generic restricted one-shot certificates (predicted column is $1/\rho(u/2)$; "curves" are aggregates over all workers at the first win).

The four certificates, in the format (p, A, x_0, m) — each verified True by `voneshot.py` and stored in `certs/restricted/` of [OSF]:

$p = 10^{20}{+}39$: `A = 19352331390561983207, x0 = 80662701055625636343, m = 14084161000`
$p = 10^{30}{+}57$: `A = 640307721818372374714786296071, x0 = 821106379164368261893854100371, m = 1341825306264338`
$p = 10^{40}{+}121$: `A = 9206317171817454415338197777434159186965, x0 = 1798766695016208561785106447263414920568, m = 113982529425281128058`
$p = 10^{50}{+}151$: `A = 52854763757971328428601223341553948579729822167, x0 = 41413023321438028880723015756658658164929271447635, m = 1037236808692880954316316 2`

The observed-over-predicted ratios $(18, 1.1, 6, 3)$ show the same order-of-magnitude scatter as the §4.3.3 race densities — single-draw variance dominates at these pool sizes. The 256-bit case ($\approx 3 \times 10^7$ candidates at ≈5–10 core-seconds each) is a 5–10 core-year project — a day or two of wall time on a ∼3000-core fleet with stock SEA: feasible, unattempted; 384 bits is out of reach. For comparison, the *unrestricted* one-shot search handled 256 bits in 4 seconds (§4.3.3): the two exponents 1/8 vs 1/4 are separated by exactly this practical chasm.

6. SHORT ECPP

6.1. **The problem.** The ShortPrimalityProofs repository [Short] adds the one ingredient the one-shot format forbids — recursion — while keeping everything else as small as possible.

Definition 6.1 ([Short]). A *short ECPP* is a sequence of integers

$$(p_0,\, A_0,\, x_0,\, m_0p_1,\, A_1,\, x_1,\, m_1p_2,\, \ldots,\, A_k,\, x_k,\, m_kp_{k+1})$$

with $k \geq 0$, $p_0 \geq 5$, and $n = \lceil \log_2 p_0 \rceil$ *fixed for the whole chain*, in which: the p_i are odd with $n^2 < p_{i+1} < \sqrt{p_i}$ for $i < k$ and $p_{k+1} = 1$; each m_i is n^2-smooth with $L_i < m_ip_{i+1} < r_iL_i$, where $L_i = L(p_i)$ and r_i is the least prime divisor of m_i; and (x_i, y_i) is a point of order m_ip_{i+1} on $B_iy^2 = x^3 + A_ix^2 + x$ *modulo every prime divisor of* p_i.

Soundness is downward induction on Lemma 2.1: $p_{k+1} = 1$ makes the last level's order fully n^2-smooth (a restricted one-shot certificate for p_k); each earlier level's order $m_ip_{i+1} > L_i$ proves p_i prime *given* that p_{i+1} is prime. Note $m_i \geq 2$ always (else the order would be $p_{i+1} < \sqrt{p_i} < L_i$), so r_i is well defined; and the certificate stores only the *product* m_ip_{i+1} — the verifier recovers p_{i+1} as its n^2-rough part by trial division below n^2, and never needs a primality test on it (the next level is the primality proof). The "modulo every prime divisor" phrasing is essential and is exactly the point of the CRT counterexample of Remark 2.2, which was found in the course of pinning this definition down.

Structure of the chain. $p_{i+1} < \sqrt{p_i}$ at least halves the bit length per level, and $p_{i+1} > n^2$ forces termination once $p_i \leq n^4$: so $k = O(\log n) = O(\log\log p_0)$ levels, and the total size is $\sum_i (2 + \frac{1}{2})\,\mathrm{bits}(p_i) \lesssim 5n$ bits (published chains measure $3.2n$–$4.7n$, the ratio rising with depth toward the halving-descent bound) — *linear* in $\log p_0$, like a Pomerance triple, against the quadratic size of a classical chain. Existence for every prime $p_0 \geq 5$ is again inherited from Pomerance triples (take $k = 0$, $m_0 = 2^\kappa$); the repository verifies it exhaustively below 2^8 (201,072 short ECPPs; every prime $5 \leq p \leq 2^8$ admits one, while $p = 2, 3$ admit none, which is why the format takes $p_0 \geq 5$).

6.2. **Verification: Pomerance-order bit complexity.** The reference verifier `vsmallECPP.py` processes the levels in order, working modulo p_i at level i (write $b_i = \lceil \log_2 p_i \rceil$, so $b_{i+1} \leq \lceil b_i/2 \rceil$ and $\sum_i b_i < 2n + O(\log n)$; one sieve to n^2 serves the whole chain since n is fixed at the top): it recovers m_i and $p_{i+1} = o_i/m_i$ by trial division of the carried product o_i over the primes $\leq n^2$, checks the window, minimality, and descent $p_{i+1}^2 < p_i$, checks that $[o_i](x_i{:}1)$ is a genuine point at infinity, and establishes exactness of the order by the §4.2 divide-and-conquer with p_{i+1} participating as one of the "primes" (its primality being the next level's job). For the accounting it pays to organize the exactness checks so that *the big prime is peeled first*: compute $U = [m_i]P$ and check $\gcd(Z_U, p_i) = 1$ — this *is* the (o_i/p_{i+1})-cofactor condition — then check $[p_{i+1}]U = O$, then run the divide-and-conquer only over the primes of m_i, applied to $V = [p_{i+1}]P$. The tree depth then multiplies only the filler's bits, never the big prime's. (The reference implementation keeps p_{i+1} inside the tree; for bounded k_i this changes nothing, and in general it costs at most one extra factor $\log k_i$ on the b_i term.)

Proposition 6.2 (verification cost, bit operations). *Let a valid short ECPP for an n-bit p_0 have levels $i = 0, \ldots, k$ with fillers m_i having k_i distinct prime factors. With the primes below n^2 and their product tree precomputed — a certificate-independent table depending only on n, built once in $O(n^2(\log n)^2)$ bit operations and stored in $O(n^2)$ bits — the certificate is verified in*

$$O\Big(\sum_i \big[\, b_i + \log m_i \cdot (1 + \log k_i) \,\big]\, M(b_i) \;+\; M(n^2)\Big)$$

bit operations: per level, $O(b_i)$ ladder bits for $[o_i]P$ and the peeled big-prime checks plus $O(\log m_i)$ ladder bits on each of the $\lceil \log_2 k_i \rceil$ levels of the exactness tree, each ladder bit costing $O(1)$ multiplications mod p_i*; globally, one batched trial division of all the o_i against the prime table at $M(n^2) = O(n^2 \log n)$* [Ber04] *(the sieve itself is $O(n^2 \log\log n)$). Consequently, whenever the fillers satisfy* $\log m_i \cdot \log k_i = O(b_i)$ *— in particular whenever $m_i \leq n^{O(1)}$, which holds for every certificate in the challenge table (the largest filler there has 50 bits, and the largest exponent* $\log m_i / \log n$ *is*

6.11*; see Remark 6.3), and also whenever every filler is a prime power, whatever its size ($k_i = 1$ makes the exactness tree trivial)* — verification costs $O(n^2 \log n)$ bit operations: the same order as verifying a Pomerance triple, exceeded only by a constant factor. *The second clause is what makes the bound attainable for* every *prime: the zero-level certificate with $m_0 = 2^\kappa \approx \sqrt{p}$ provided by Pomerance's existence argument (§3.2) has $k_0 = 1$. (Polynomially bounded fillers are what every* found *certificate has; no argument guarantees a full chain of them for every prime.)*

The constant is roughly an order of magnitude with the reference formulas: the two-register ladder spends 12 multiplications per scalar bit against the doubling-only chain's 5, the level sum $\sum b_i$ approaches $2n$ against Pomerance's single $(n/2)$-bit chain, and the trial-division pass adds $\approx M(n^2) \approx 2n^2 \log_2 n$. In the format's worst case — a filler of $\Theta(b_i)$ bits with $\Theta(b_i/\log b_i)$ distinct primes, which the definition permits and which is exactly what the $k = 0$ stratum (= the restricted one-shot, §5) can look like — the tree factor is genuine and verification degrades to $\Theta(n^2(\log n)^2)$, the one-shot bound, as the inclusion of §5 forces. Both statements sharpen the repository's $O((\log p_0)^{2+o(1)})$; the chain structure — one big prime per level, geometrically shrinking moduli, small smooth fillers — is precisely what deletes the one-shot's extra log factor, the same mechanism as in the fixed-ratio chains of §6.5.

Remark 6.3 (tightened specifications with worst-case $O(n^2 \log n)$ verification). The two potential sources of a second log factor in Proposition 6.2 are distinct in kind. (a) *The exactness tree*: its depth $\log k_i$ multiplies the filler bits, and the definition permits a filler of $\Theta(b_i)$ bits with $\Theta(b_i/\log b_i)$ distinct primes, for which the top level costs $\Theta(n^2(\log n)^2)$ — a worst case over certificates, vacuous for the ones the finder produces. (b) *The trial division*: the per-certificate cost is $M(n^2) = \Theta(n^2 \log n)$ — Pomerance-order — but only given the precomputed product tree of the primes below n^2, whose construction costs $\Theta(n^2(\log n)^2)$; a self-contained accounting carries the extra log here instead.

Capping the fillers. Requiring only $m_i \le n^c$ for a fixed c, with the certificate payload unchanged, removes both sources — though one must be careful where the trial division happens. Direct per-prime trial division of o_i costs $\Omega(n_i^3/\log n_i)$ at a level of n_i bits — each of the $\pi(n_i^2) \approx n_i^2/(2\ln n_i)$ candidate primes must scan the $(n_i/2)$-bit o_i, $\Omega(n_i)$ apiece — a full factor $n_i/(\log n_i)^2$ above the target. Instead the verifier splits the recovery in two stages. First, extract the smooth parts of *all* levels in one batched pass: reduce the primorial $P = \prod_{q \le n^2} q$ modulo $\prod_i o_i$ at cost $M(1.44\, n^2) = O(n^2 \log n)$, remainder-tree down to the individual o_i ($\widetilde{O}(n)$), and apply the repeated-squaring gcd trick at each level ($\widetilde{O}(n_i)$). Second, factor each extracted $m_i \le n^c$ — now a $c \log_2 n$-bit integer — directly: trial division to n for its small primes ($\widetilde{O}(n)$), then deterministic Pollard–Strassen splitting [Pol74, Str77] of the $\le c$ remaining factors, using the format's smoothness bound: every prime factor of m_i lies below n^2, so the search radius is $B = n^2$ and each split costs $\widetilde{O}(\sqrt{B}) = \widetilde{O}(n)$ — the generic $\widetilde{O}(m_i^{1/4}) = \widetilde{O}(n^{c/4})$ bound is never invoked, and the stage is $\widetilde{O}(n)$ per level *for every fixed c*. Everything after the single batched reduction thus costs $\widetilde{O}(n)$ per level (the elliptic work summing geometrically, the factoring work adding $\widetilde{O}(n)$ across the $O(\log n)$ levels), so verification is worst-case $O(n^2 \log n)$ — with the one caveat that the primorial, a single $1.44\, n^2$-bit integer depending only on n (about 80 KB at 665 bits), is a per-bit-size precomputation whose cold construction still costs $\Theta(n^2(\log n)^2)$. A terminal level should then be permitted only once $\sqrt{p_k} \le n_k^c$, pushing the chain one polylogarithmically cheap level deeper. The exactness checks collapse as in Proposition 6.2: one half-length ladder $V = [p_{i+1}]P$ per level plus scalars $\le n^c$, for $\sum_i (b_i/2 + O(\log^2 n))$ ladder bits and a constant roughly $5\times$ Pomerance's.

Empirical compliance. Against the twenty published chains (70 levels): keeping the smoothness bound at the top-level n^2, as in the current definition, a cap with $c = 7$ retains every certificate — the extremal fillers are 46 bits at 10^{50}+151 (level 0, $m = n^{6.11}$) and 50 bits at 10^{90}+289 (level 0, $n^{5.97}$). Re-indexing the bounds per level ($n_i = \lceil \log_2 p_i \rceil$, fillers n_i^2-smooth and $\le n_i^c$) is the cleaner specification asymptotically but is not backward compatible: 23 of the 70 levels, hitting 17 of the 20 chains, contain a filler prime exceeding n_i^2 (deep and terminal levels were built under the top-level bound), and the per-level cap would need $c \geq 6.5$. Either way the finder is essentially unaffected: the $L(1/3)$ channel's fillers are smooth parts of typical size $n_i^{2+o(1)}$, and a cap only truncates a thin tail of lucky large-filler winners — a constant-factor effect, not an asymptotic one.

Factored fillers. If one is willing to change the payload instead, carrying each filler in factored form removes even the primorial: the verifier checks each listed prime $q_j \leq n^2$ by trial division to $\sqrt{q_j} \leq n$, and the n^2-roughness of p_{i+1} need never be checked at all — a composite p_{i+1} fails at the next level, by the same induction that already carries the soundness proof. Verification is then worst-case $O(n^2 \log n)$ with $O(n)$ memory and *no* precomputation of any kind, at the price of certificates that are no longer unique for a given (p_i, A_i, x_i, o_i).

The uncapped format, all-inclusive. Conversely, if one keeps the current definition unchanged, the worst case is $O(n^2(\log n)^2)$ *with the per-n precomputation included*: the primorial product tree costs $\Theta(n^2(\log n)^2)$ to build, one reduction against $\prod_i o_i$ costs $M(n^2) = O(n^2 \log n)$, the shared active-branch descent delivering every level's prime divisors costs $O(n^2 \log n)$ more, and the elliptic worst case (a top-level filler of $\Theta(n)$ bits with $\Theta(n/\log n)$ primes) is $\Theta(n^2(\log n)^2)$ exactly — as it must be, since the $k = 0$ stratum is the restricted one-shot. The one proviso is organizational: the trial division must be batched *across the whole chain* (build the tree once, reduce it once against $\prod_i o_i$, share the descent); repeating the n^2-scale pass level by level costs an extra logarithmic factor. (The reference verifier currently does repeat the pass per level, and additionally builds its batch products by sequential multiplication, which is quadratic in the batch size — $\widetilde{\Theta}(n^3)$ asymptotically, though with so small a constant that it is invisible below $n \sim 10^6$; both are local changes to the implementation, not to the format. Patched verifiers realizing the batched organization, with unchanged accept/reject behavior, accompany [OSF] in `verifier-batching-pr/`.) The three regimes, then: the current definition verifies in $O(n^2(\log n)^2)$ fully self-contained; capped fillers give $O(n^2 \log n)$ plus a once-per-n $O(n^2(\log n)^2)$ table; factored fillers give $O(n^2 \log n)$ outright.

One formula in fact covers the elliptic work of the whole family of §§3–6: $O(\sum_i [\log o_i + \log m_i(1 + \log k_i)]\, M(b_i))$ bit operations, where o_i is the certified point order at level i and m_i its smooth part with k_i distinct primes. A Pomerance triple is one level with $\log m = n/2$ and $k = 1$: $O(n\, M(n)) = O(n^2 \log n)$. A restricted one-shot has $\log m = n/2$ with k free: Pomerance-order for bounded k, and $\Theta(n^2(\log n)^2)$ at $k = \Theta(n/\log n)$. The n^4 one-shot is the same with its listed primes included in k. A short ECPP keeps every $\log m_i$ small and pushes all the large bits into the once-per-level big prime — which is why it alone combines subexponential findability with Pomerance-order verification.

6.3. **Finding certificates: the free prime makes it subexponential.** The search problem at one level of the chain: given p (a level modulus), find a curve and a divisor $o = mq$ of $\#E$ with m n^2-smooth, q prime in $(n^2, \sqrt{p})$, and o in the window (L, rL). Compare §5: the order may now contain *one arbitrary prime* up to $\sqrt{p}$ — but the prover must *discover* q, i.e. partially factor $\#E$. The reference finder `short.gp` [Short] (SEA on random curves and their twists, so two orders per point count) recognizes winners through three channels of increasing cost:

1. *Terminal:* some divisor of the n^2-smooth part s of $\#E$ lands in the window — a restricted one-shot certificate; ends the chain ($p_{\text{next}} = 1$). Density as in §5: $\rho(u/2)$ with $u = \ln p / \ln n^2$.
2. *Cheap prime rough part:* the rough cofactor $R = \#E/s$ is itself prime and below $\sqrt{p}$. Costs one pseudoprimality test, but demands that the entire discarded half of $\#E$ be n^2-smooth: measured density 25% of curves at 10^{10}, 0 of 200 by 10^{40} — this channel alone does not scale.
3. *General descent (the workhorse):* factor R under a time budget (`factorint` inside an `alarm`; the budget is an early abort that keeps the search on curves whose orders factor

easily) and use any exposed prime $q \in (n^2, \sqrt{p})$ with a smooth filler $m \mid s$ completing the window. The discarded cofactor $\#E/o$ is unconstrained — this is what channel (2) lacked.

Unlucky sub-chains are abandoned by bounded backtracking (a descent gets a budget of 64 curves before the parent resamples), and `parallel_short.py` races independent workers mixing complete- and partial-factorization strategies. At small levels complete factorization is available and the finder descends faster than square root (observed chains like $665 \to 333 \to 159 \to 47$ bits).

Proposition 6.4 (heuristic finding cost)**.** *Fix $b \in [\frac{1}{2}, 1)$ and put $\sigma = 1 - b$. Under the heuristics (H1)–(H3) below, the budgeted-ECM search above (channels (1)–(3)) finds one level of a ratio-b chain at modulus p — in particular a short ECPP level, $b = \frac{1}{2}$ — at expected cost*

$$L_p\Big(1/3,\ (3\sigma)^{1/3}\Big) \ = \ L_p\Big(1/3,\ (3/2)^{1/3} \approx 1.14\Big) \quad \textit{at } \sigma = \tfrac{1}{2},$$

attained with ECM budget $y = L_p\Big(2/3,\ (\sigma^2/3)^{1/3}\Big)$, and the same for both order supplies: SEA point counts contribute only polynomial factors, while the CM scan and winner class polynomial cost $L_p\Big(1/3,\ (8\sigma/9)^{1/3}\Big)$ — exponent coefficient exactly two thirds of the total — and hence never dominate. The level exponents scale by $2^{-i/3}$ down the chain, so the top level dominates and the whole chain costs the same: subexponential, with a smaller constant than the $L_p\Big(1/3,\ (64/9)^{1/3} \approx 1.92\Big)$ of factoring p itself by NFS [LL93].

Heuristic derivation. The heuristics, all from §2: *(H1)* candidate orders behave as independent uniform Hasse-interval integers (§2.4; the local biases are $O(1)$ factors, invisible on the L-scale). *(H2)* the event that a candidate $N \approx p$ splits as $N = Sq$ with q prime in the acceptance window and $S = N/q$ y-smooth has probability $\rho(u)^{1+o(1)}$ with $u = \sigma \ln p / \ln y$: winners concentrate at maximal q, i.e. at cofactor size $S \approx p^\sigma$, since a smaller q forces a larger smooth cofactor, which is doubly exponentially less likely, and the prime-density and window-width factors are polynomial in $\log p$, hence absorbed by the $o(1)$ (§2.5). *(H3)* ECM peels every prime factor $\le y$ of a candidate in expected time $L_y(1/2, \sqrt{2}) \cdot \mathrm{poly}(n)$ [Len87]; a candidate that fails to peel within the budget is discarded (the early abort of channel (3)), which affects only the search, never soundness.

These heuristics model exactly the budgeted prover, and (H2) is its *discoverability* condition, not an added restriction: the prover can exhibit q only if everything else in N peels — a second prime factor above y would sit with q in the unfactored rough part and hide it — while the discarded cofactor $N/(mq)$ is otherwise arbitrary within y-smoothness (the filler m is a *divisor* of the smooth part, not all of it). The one channel outside the model — $q \le y$, exposed by the peeling itself, with the window completed by a large filler — needs an n^2-smooth part exceeding $\sqrt{p}/y = p^{1/2-o(1)}$, an event of probability $p^{-1/4+o(1)}$ (the restricted-one-shot density, §5.2), far below the main channel's $\rho(u) = 1/L_p(1/3,\ \cdot)$ and hence negligible. Finally, the clean threshold — every prime $\le y$ peels, nothing above does — idealizes the reference finder's `factorint`-under-`alarm` budget; the soft cutoff moves constants, not the exponent.

Write $P_1 = \ln p$, $P_2 = \ln\ln p$, and $X = P_1^{1/3} P_2^{2/3}$, so that $L_p(1/3, c) = \exp((c + o(1))X)$, and take $\ln y = \gamma P_1^{2/3} P_2^{1/3}$ — the scale is forced by the balance below. Then $u = (\sigma/\gamma) P_1^{1/3} P_2^{-1/3}$ and $\ln u = \frac{1}{3} P_2 (1 + o(1))$, so the expected number of candidates is

$$1/\rho(u)^{1+o(1)} \ = \ \exp\!\big(u \ln u\,(1 + o(1))\big) \ = \ \exp\Big(\big(\tfrac{\sigma}{3\gamma} + o(1)\big)X\Big);$$

and $\ln\ln y = \frac{2}{3} P_2 (1 + o(1))$ — *not* P_2; this is where conventions matter (Remark 6.5) — gives per-candidate peeling cost

$$L_y(1/2, \sqrt{2}) \ = \ \exp\!\big((\sqrt{2} + o(1))(\ln y\ \ln\ln y)^{1/2}\big) \ = \ \exp\Big(\big(2(\gamma/3)^{1/2} + o(1)\big)X\Big).$$

The total exponent coefficient $c(\gamma) = \sigma/(3\gamma) + 2(\gamma/3)^{1/2}$ is minimized at $\gamma^* = (\sigma^2/3)^{1/3}$, where the peeling term is exactly twice the candidate-count term and $c(\gamma^*) = 3\,\sigma^{1/3}3^{-2/3} = (3\sigma)^{1/3}$.

Supplies. With SEA, each candidate adds one point count, $\mathrm{poly}(n) = \exp(o(X))$. With CM, K candidates require discriminants up to $B = \widetilde{\Theta}(K^2)$ (§2.6.2), so the scan — and the winner's class polynomial, $\widetilde{O}(B)$ — costs $\exp\big((2\sigma/(3\gamma) + o(1))X\big)$; at γ^* this coefficient is $2\sigma^{1/3}3^{-2/3} = (8\sigma/9)^{1/3}$, strictly below $c(\gamma^*)$. Nor can any other γ improve the CM total $\max\big(2\sigma/(3\gamma),\, c(\gamma)\big)$: $c(\gamma^*)$ is the global minimum of c, and the scan term sits below it there, so the minimum of the max is again $c(\gamma^*)$ — the scan-versus-peeling trade never binds. Finally, the level at the $n/2^i$-bit modulus costs $\exp\big(c\,2^{-i/3}X(1 + o(1))\big)$, a geometrically shrinking exponent, so the chain total is dominated by its top level. □

Remark 6.5 (conventions, and the previously quoted constants 1.31 and 1.39). Earlier revisions of this write-up, following the prototype analysis in [OSF] (report *descent-chain*), quoted $L_p\Big(1/3,\ (9\sigma/2)^{1/3} \approx 1.31\Big)$ for the SEA supply and $L_p\Big(1/3,\ (2/3)(18\sigma)^{1/3} \approx 1.39\Big)$ for CM. Those constants come from two coarser choices. Approximating $\ln\ln y$ by $\ln\ln p$ (rather than $\frac{2}{3}\ln\ln p$) inflates the peeling coefficient by $(3/2)^{1/2}$ and the optimized constant by $(3/2)^{1/3} \approx 1.14$, turning $(3\sigma)^{1/3}$ into $(9\sigma/2)^{1/3}$; and, for CM, imposing the balance "scan = candidates × peeling" — which the minimax in the proof shows to be non-binding — yields $(16\sigma/3)^{1/3} = (2/3)(18\sigma)^{1/3}$. The corrected accounting makes the two supplies share the leading constant; the real difference between them lives in the polynomial factors (a stock SEA count per candidate against a polylogarithmic amortized scan), which matter at feasible sizes but are invisible on the L-scale. And since all these expressions converge only at the $\ln\ln\ln p/\ln\ln p$ rate, no such constant is quantitatively meaningful at feasible sizes anyway: the robust conclusions are the exponent $1/3$, the $(1-b)^{1/3}$ shape of the b-dependence, and "cheaper than factoring p."

6.4. **Results to date.** The challenge table [Short] holds verified short ECPPs for the least prime above 10^c, $c = 10, 20, \ldots, 200$ — the top entry at 665 bits, twice the one-shot frontier, found on hardware where the deepest one-shot runs needed hundreds of cores (Table 7).

p	bits	levels	CPU time	p	bits	levels	CPU time
$10^{10}+19$	34	2	<1 s	$10^{110}+7$	366	4	≈850 s
$10^{20}+39$	67	2	≈1 s	$10^{120}+79$	399	4	≈1400 s
$10^{30}+57$	100	2	≈1 s	$10^{130}+1113$	432	3	≈55 s
$10^{40}+121$	133	3	≈2 s	$10^{140}+13$	466	4	≈800 s
$10^{50}+151$	167	3	≈3 s	$10^{150}+67$	499	5	≈1500 s
$10^{60}+7$	200	3	≈3 s	$10^{160}+303$	532	5	≈1200 s
$10^{70}+33$	233	4	≈45 s	$10^{170}+63$	565	4	≈830 s
$10^{80}+129$	266	4	≈180 s	$10^{180}+313$	599	4	≈850 s
$10^{90}+289$	299	3	≈350 s	$10^{190}+253$	632	4	≈1400 s
$10^{100}+267$	333	3	≈800 s	$10^{200}+357$	665	4	≈4400 s

TABLE 7. The ShortPrimalityProofs challenge table [Short]. Entries through $c = 100$ by `short.gp` on a single core (AVS and Claude Code, Fable 5); $c = 110, \ldots, 200$ by `parallel_short.py` workers.

The growth from 3 s at 200 bits to 4,400 s at 665 bits — a factor $\approx 10^3$ over a ×3.3 size increase, with large per-draw variance (the lucky 432-bit chain cost 55 s) — is consistent with the gentle $L(1/3)$ curve; extrapolating, 10^{300} (997 bits) is a core-days-to-weeks project.

6.5. **Interpolating toward classical ECPP: fixed-ratio descent chains.** Relaxing the descent rate from "square root" to a parameter: a chain in which each step's certified prime satisfies $q \le p^b$

(and the point order is just $c{\cdot}q$ with no minimality demanded) was analyzed and prototyped in [OSF] (report *descent-chain*) for $b \in (\frac{1}{2}, 1)$: certificate size $O(n)/(1-b)$; verification $O(n^2 \log n)/(1-b^2)$ bit operations — Pomerance's order, the geometric shrink killing the one-shot's extra log factor; finding $L_p\Big(1/3,\ (3(1-b))^{1/3}\Big)$ (Proposition 6.4; the prototype report's original constants used the coarser convention of Remark 6.5), which tends to 0 as $b \to 1$, at the price of $\Theta(n)$ levels and quadratic certificates. The interpolation toward classical ECPP is qualitative only: the estimate's $o(1)$ terms are not uniform in b, so letting $b = b(n) \to 1$ does not *recover* the polynomial-time bound — at polylogarithmic cofactor sizes the Dickman analysis gives way to the prime-density analysis of §7. Short ECPP is the $b = \frac{1}{2}$ endpoint with minimized orders. Measured with $b = 0.85$ on a 16-core desktop: $2^{255} - 19$ proved in 3.7 s (vs ≈15 s one-shot), $10^{100} + 267$ in 3.7 s (vs 7.9 min one-shot), 1024 bits in 128 s (9 steps, verification 405 ms), 2048 bits in 5.2 h (12 steps, verification 2.65 s) — sizes far beyond any one-shot's reach, with verification still two orders of magnitude cheaper than a classical chain's at equal size.

7. Classical ECPP

7.1. **The problem and the two classical algorithms.** The classical elliptic curve primality proof drops all structural demands on the order: a chain $p = p_0 > p_1 > \cdots$, where one step exhibits, for $p = p_i$, a curve $E/\mathbb{F}_p$ with $\#E = c \cdot q$, q a probable prime exceeding $L(p)$, together with a point P such that $Q = [c]P \neq O$ and $[q]Q = O$; the next level certifies $q = p_{i+1}$. By Lemma 2.1 (applied to the point Q of prime order $q > L(p)$ modulo every prime divisor) each step is sound, and the chain bottoms out at a size testable by trial division. The exhibited order is a single prime; the cofactor c needs no factorization and carries no burden of proof — its peelability is purely the prover's recognition problem. Two ways to run a step:

- **SEA supply (Goldwasser–Kilian [GK86]; the "basic" route):** sample random curves over $\mathbb{F}_p$, count points, and keep a curve with $\#E = 2q$, q a probable prime ($\approx p/2$, so the chain sheds one bit per step). Expected $O(\log p)$ curves per step by the prime number theorem *if* the Hasse interval contains the expected number of primes — provable for all inputs outside a set of primes of relative density zero using unconditional results on primes in short intervals [HB88], conjecturally for all. With SEA at $\widetilde{O}((\log p)^4)$ per count and $\Theta(\log p)$ curves per step, a step costs $(\log p)^{5+\varepsilon}$, and the downrun — $\Theta(\log p)$ steps for the strict $m = 2q$ descent, a logarithmic factor fewer with polynomially smooth cofactors peeled — puts the route at $(\log p)^{6+o(1)}$ in total. Adleman and Huang [AH92] closed the gap of exceptional primes by a random reduction through Jacobians of genus-2 curves, putting primality in ZPP — a theorem, though never a practical algorithm; their certificates involve two-dimensional abelian varieties, so this does not give every prime a certificate in the elliptic-curve format defined above.
- **CM supply (Atkin–Morain [AM93]; fastECPP):** replace point counting by the CM supply of §2.6.2: for each small discriminant D with $4p = t^2 + |D|v^2$, the candidate orders $p + 1 \pm t$ are known before any curve is built; trial-divide them, keep one of the form $c \cdot q$ with c smooth and q a probable prime, and only then construct the curve from a root of H_D. Because tiny discriminants suffice (the step needs only *some* usable order, an event of probability $\approx 1/\log p$ per candidate), the class polynomials stay small, and the certified prime can be pushed down by the cofactor's length (typically $\Theta(\log n)$ bits per step, giving $\Theta(n/\log n)$ to $\Theta(n)$ steps). Heuristic cost $(\log p)^{5+\varepsilon}$ for the basic version — dominated by the per-discriminant square roots — and $(\log p)^{4+\varepsilon}$ for *fastECPP* [FKMW04, Mor07], which batches the square roots $\sqrt{D} \bmod p$ across all D via one square root per small prime discriminant factor (Shallit's trick): the fastest known heuristic primality *proving*

complexity for general p. Enge's MPI implementation [Eng24] parallelizes it to quasi-cubic wall-clock time on a linear number of cores.

7.2. **Certificates and verification.** A classical certificate stores, per step, $(p_i, \mathrm{curve}, m_i, q_{i+1}, P_i)$: ≈ 5 integers of b_i bits with b_i shrinking only $\theta = \Theta(\log n)$ per step, so $\Theta(n^2/\log n)$ bits in all. Verification re-runs the two scalar multiplications of every step: $\sum_j O(b_j)\, M(b_j)$ over $\Theta(n/\theta)$ steps of linearly decaying size is $O(n^3 \log n/\theta) = O(n^3)$ bit operations for the customary cofactor lengths — the M-factor cancels against the step count — quasi-cubic, the benchmark against which the $O(n^2 \log n)$-class formats of §§3–6 are the improvement. These are typical-output figures for the customary smooth-cofactor regime, not worst cases over the format: the strict Goldwasser–Kilian descent $m = 2q$ sheds only one bit per step ($\theta = 1$), giving $\Theta(n^2)$ bits and $O(n^3 \log n)$ verification, and the format equally admits (rare, never searched for) steeper chains that would verify faster. Table 1 quotes the customary regime. (Verification is embarrassingly parallel across steps, a fact used in practice for the record certificates below.)

7.3. **Results to date.** Classical ECPP is the workhorse of general-purpose primality proving, with three decades of records: from the sub-1,000-digit proofs of the first ECPP implementations [AM93] through $\approx 10^4$–2×10^4 digits in the fastECPP era of the mid-2000s [FKMW04, Mor07], to today's frontier held by Enge's CM/fastECPP-over-MPI [Eng24] and Marcel Martin's Primo (the standard desktop tool). The largest ECPP-proven primes as of mid-2026 [t5k]:

prime	digits	date	notes
$R(109297) = (10^{109297} - 1)/9$	109,297	May 2025	repunit; A. Enge & P. Underwood, fastECPP/MPI
$R(86453)$	86,453	May 2023	repunit; previous record
$5^{104824} + 1048245$	73,269	Feb 2023	
$(2^{221509} - 1)/292391881$	66,673	Oct 2023	Mersenne cofactor

For perspective across the whole write-up: at 10^5 digits the *only* practical route is the classical one — every other format in this survey buys its cheaper verification with a finding cost that is subexponential-to-exponential in the bit length and hence confined to hundreds of bits (one-shot) or $\approx 10^3$ bits (short/descent chains). Conversely, at the sizes where they do apply, the §3–6 formats verify orders of magnitude faster than a classical chain, with certificates orders of magnitude smaller.

8. Synthesis and open problems

The five formats populate a two-parameter design space: *how much order structure is demanded* (a power of two; n^2-smooth; n^4-smooth with listed witnesses; smooth × one-prime; a single free prime) and *how much recursion is allowed* (none; square-root descent; ratio-b descent; free descent). Moving along either axis trades finding cost against verification cost and certificate size, and Table 1 summarizes the endpoints. The empirical record across all four challenge repositories matches the heuristic model remarkably well: candidate counts within a factor 1–20 of the Dickman/counting predictions at every measured size, with the systematic part of the discrepancy accounted for by known local biases and structure conditions (§2.4), and single-draw geometric variance dominating the rest.

Open problems, in roughly increasing order of depth:

(1) *Find a Pomerance triple for* $10^{27} + 103$ [DANGER3] — the standing challenge; $\approx 3 \times 10^{13}$ mean-case candidates at the measured GPU rate (5×10^7 per second) is around 170 GPU-hours.

(2) *Beat* $\widetilde{\Theta}(\sqrt{p})$ *for Pomerance triples*, or give evidence none of the standard toolkits (CM, isogenies, torsion families, sieving) can. The obstruction is that "2^k divides $\#E$" is an event of probability $2^{-n/2+o(n)}$ with no multiplicative structure to sieve on (§3.4).

(3) *Close the one-shot $p^{1/8}$ vs $p^{1/4}$ gap.* Two separate barriers hold the CM route at $p^{1/4+o(1)}$ (§4.3.2): the scan costs $\widetilde{\Theta}(B)$ to find the usable (D,t,v) with $4p = t^2 + |D|v^2$ below B, and the winner's class polynomial costs $\widetilde{O}(|D|) = \widetilde{O}(B)$ — with $B \approx K^2$ forced by the $1/\sqrt{|D|}$ solvability density, not by the algorithm. Closing the gap needs both an output-sensitive enumeration of usable (D,t,v) and a commit cheaper than quasi-linear-in-$|D|$ class polynomial construction (say, one root of H_D modulo p in $h^{1+o(1)} = |D|^{1/2+o(1)}$ time — a long-standing open problem in its own right); either alone still leaves $p^{1/4+o(1)}$, and both together would move the SEA crossover (§4.3.3) beyond reach.

(4) *Instrument SEA for early-abort supply*: expose the order modulo small ℓ during the Elkies phase to gate candidates before the full $\widetilde{O}(n^4)$ count (modeled $\times$3–5 at one-shot sizes; needs internals no current library exports).

(5) *A generic 256-bit restricted certificate* (§5.3): $\approx 3 \times 10^7$ SEA candidates — 5–10 core-years, days on a ∼3000-core fleet with stock tools, much less with item 4.

(6) *Rigorous existence with the right densities.* Existence of all four quasi-quadratic formats is unconditional (via [Pom87]), but every *finding* bound in this write-up is heuristic; even the classical chain's polynomial time is only "almost all p" provable [GK86, HB88] (all p in expected polynomial time only via [AH92]). Proving that a positive proportion of curves over every $\mathbb{F}_p$ has $S_y(\#E) > \sqrt{p}$ for the relevant y would derandomize a whole column of the table.

(7) *Lower bounds.* Pomerance's discussion [Pom87, §1] still delimits what is known: some primes (Fermat, Mersenne) have $(1 + o(1)) \log_2 p$-multiplication proofs; nothing rules out $O(\log \log p)$ multiplications for special primes; and whether $(5/2) \log_2 p$ is optimal for general p — he declined to conjecture it — remains untouched. The formats here show the *finding* cost of the known constructions rising steeply as verification approaches his constant; whether that trade-off is intrinsic is open.

(8) *The find/verify trade-off.* Exhibit a certificate family achieving both finding in (even randomized, even heuristic) polynomial time and verification in $O(n^{2+o(1)})$ bit operations. (Certificate size needs no third demand: a verifier must read what it uses, so effective size never exceeds verification time — and the AKS–Bernstein family shows a small certificate can even be worthless, Remark 1.1.) Short ECPP achieves the verification side with $L_p(1/3, \approx 1.14)$ finding; the AKS–Bernstein certificate achieves the finding side — expected *quadratic* time — but verifies quartically, for every known way of checking its central congruence (Remark 1.1). No known family achieves both.

References

[AH92] L. M. Adleman, M.-D. A. Huang, *Primality Testing and Abelian Varieties over Finite Fields*, Lecture Notes in Math. 1512, Springer, 1992, doi:10.1007/BFb0090185.

[AKS04] M. Agrawal, N. Kayal, N. Saxena, PRIMES is in P, *Ann. of Math.* 160 (2004), 781–793, doi:10.4007/annals.2004.160.781.

[AM93] A. O. L. Atkin, F. Morain, Elliptic curves and primality proving, *Math. Comp.* 61 (1993), 29–68, doi:10.1090/S0025-5718-1993-1199989-X.

[Ber04] D. J. Bernstein, How to find smooth parts of integers, preprint, 2004, `https://cr.yp.to/papers.html#smoothparts`.

[Ber07] D. J. Bernstein, Proving primality in essentially quartic random time, *Math. Comp.* 76 (2007), 389–403, doi:10.1090/S0025-5718-06-01786-8.

[Ber98] D. J. Bernstein, Detecting perfect powers in essentially linear time, *Math. Comp.* 67 (1998), 1253–1283, doi:10.1090/S0025-5718-98-00952-1.

[Berr05] P. Berrizbeitia, Sharpening "PRIMES is in P" for a large family of numbers, *Math. Comp.* 74 (2005), 2043–2059, doi:10.1090/S0025-5718-05-01727-8.

[BLS75] J. Brillhart, D. H. Lehmer, J. L. Selfridge, New primality criteria and factorizations of $2^m \pm 1$, *Math. Comp.* 29 (1975), 620–647, doi:10.1090/S0025-5718-1975-0384673-1.

[CEP83] E. R. Canfield, P. Erdős, C. Pomerance, On a problem of Oppenheim concerning "factorisatio numerorum," *J. Number Theory* 17 (1983), 1–28, doi:10.1016/0022-314X(83)90002-1.

[Cor08] G. Cornacchia, Su di un metodo per la risoluzione in numeri interi dell'equazione $\sum C_h x^{n-h} y^h = P$, *Giornale di Matematiche di Battaglini* 46 (1908), 33–90.

[Cox13] D. A. Cox, *Primes of the Form $x^2 + ny^2$*, 2nd ed., Wiley, 2013, doi:10.1002/9781118400722.

[CP05] R. Crandall, C. Pomerance, *Prime Numbers: A Computational Perspective*, 2nd ed., Springer, 2005, doi:10.1007/0-387-28979-8.

[Deu41] M. Deuring, Die Typen der Multiplikatorenringe elliptischer Funktionenkörper, *Abh. Math. Sem. Univ. Hamburg* 14 (1941), 197–272, doi:10.1007/BF02940746.

[Dic30] K. Dickman, On the frequency of numbers containing prime factors of a certain relative magnitude, *Ark. Mat. Astr. Fys.* 22A (1930), 1–14.

[Elk98] N. D. Elkies, Elliptic and modular curves over finite fields and related computational issues, in *Computational Perspectives on Number Theory*, AMS/IP Stud. Adv. Math. 7 (1998), 21–76.

[Eng09] A. Enge, The complexity of class polynomial computation via floating point approximations, *Math. Comp.* 78 (2009), 1089–1107, doi:10.1090/S0025-5718-08-02200-X.

[Eng24] A. Enge, FastECPP over MPI, 2024, arXiv:2404.05506; software CM/cm-ecpp, `https://www.multiprecision.org/cm/ecpp.html`.

[FKMW04] J. Franke, T. Kleinjung, F. Morain, T. Wirth, Proving the primality of very large numbers with fastECPP, in *Algorithmic Number Theory (ANTS-VI)*, LNCS 3076, Springer, 2004, doi:10.1007/978-3-540-24847-7_14.

[FM02] M. Fouquet, F. Morain, Isogeny volcanoes and the SEA algorithm, in *Algorithmic Number Theory (ANTS-V)*, LNCS 2369, Springer, 2002, 276–291, doi:10.1007/3-540-45455-1_23.

[GK86] S. Goldwasser, J. Kilian, Almost all primes can be quickly certified, in *Proc. 18th STOC*, 1986, 316–329, doi:10.1145/12130.12162; journal version: Primality testing using elliptic curves, *J. ACM* 46 (1999), 450–472, doi:10.1145/320211.320213.

[Gra08] A. Granville, Smooth numbers: computational number theory and beyond, in *Algorithmic Number Theory*, MSRI Publ. 44, Cambridge, 2008, 267–323.

[HB88] D. R. Heath-Brown, The number of primes in a short interval, *J. reine angew. Math.* 389 (1988), 22–63, doi:10.1515/crll.1988.389.22.

[Hil86] A. Hildebrand, On the number of positive integers $\leq x$ and free of prime factors $> y$, *J. Number Theory* 22 (1986), 289–307, doi:10.1016/0022-314X(86)90013-2.

[HvdH21] D. Harvey, J. van der Hoeven, Integer multiplication in time $O(n \log n)$, *Ann. of Math.* 193 (2021), 563–617, doi:10.4007/annals.2021.193.2.4.

[Koh96] D. Kohel, *Endomorphism rings of elliptic curves over finite fields*, PhD thesis, University of California, Berkeley, 1996.

[Len87] H. W. Lenstra, Jr., Factoring integers with elliptic curves, *Ann. of Math.* 126 (1987), 649–673, doi:10.2307/1971363.

[LL93] A. K. Lenstra, H. W. Lenstra, Jr. (eds.), *The Development of the Number Field Sieve*, Lecture Notes in Math. 1554, Springer, 1993, doi:10.1007/BFb0091534.

[LP19] H. W. Lenstra, Jr., C. Pomerance, Primality testing with Gaussian periods, *J. Eur. Math. Soc.* 21 (2019), 1229–1269, doi:10.4171/JEMS/861.

[McL26] A. McLain (with GPT 5.5 Codex / Claude), Danger 2026 Data Challenge code and research notes, 2026, `https://github.com/alexamclain/Danger2026DataChallenge`.

[Mil76] G. L. Miller, Riemann's hypothesis and tests for primality, *J. Comput. System Sci.* 13 (1976), 300–317, doi:10.1016/S0022-0000(76)80043-8.

[Mon87] P. L. Montgomery, Speeding the Pollard and elliptic curve methods of factorization, *Math. Comp.* 48 (1987), 243–264, doi:10.1090/S0025-5718-1987-0866113-7.

[Mor07] F. Morain, Implementing the asymptotically fast version of the elliptic curve primality proving algorithm, *Math. Comp.* 76 (2007), 493–505, doi:10.1090/S0025-5718-06-01890-4.

[Pol74] J. M. Pollard, Theorems on factorization and primality testing, *Proc. Cambridge Philos. Soc.* 76 (1974), 521–528, doi:10.1017/S0305004100049252.

[Pom87] C. Pomerance, Very short primality proofs, *Math. Comp.* 48 (1987), 315–322, doi:10.1090/S0025-5718-1987-0866117-4.

[Pra75] V. R. Pratt, Every prime has a succinct certificate, *SIAM J. Comput.* 4 (1975), 214–220, doi:10.1137/0204018.

[Rab80] M. O. Rabin, Probabilistic algorithm for testing primality, *J. Number Theory* 12 (1980), 128–138, doi:10.1016/0022-314X(80)90084-0.

[Rue26] F. Ruehle (with Claude Opus 4.6), Danger 2026 Data Challenge implementation, 2026, `https://github.com/ruehlef/Danger2026DataChallenge`; see also J. Shi, `https://github.com/JaneShi99/pomerance-p24`.

[Sch85] R. Schoof, Elliptic curves over finite fields and the computation of square roots mod p, *Math. Comp.* 44 (1985), 483–494, doi:10.1090/S0025-5718-1985-0777280-6.

[Sch87] R. Schoof, Nonsingular plane cubic curves over finite fields, *J. Combin. Theory Ser. A* 46 (1987), 183–211, doi:10.1016/0097-3165(87)90003-3.

[Sch95] R. Schoof, Counting points on elliptic curves over finite fields, *J. Théor. Nombres Bordeaux* 7 (1995), 219–254, doi:10.5802/jtnb.142.

[Sil09] J. H. Silverman, *The Arithmetic of Elliptic Curves*, 2nd ed., GTM 106, Springer, 2009, doi:10.1007/978-0-387-09494-6.

[SS14] I. E. Shparlinski, A. V. Sutherland, On the distribution of Atkin and Elkies primes, *Found. Comput. Math.* 14 (2014), 285–297, doi:10.1007/s10208-013-9181-9.

[Str77] V. Strassen, Einige Resultate über Berechnungskomplexität, *Jahresber. Deutsch. Math.-Verein.* 78 (1976/77), 1–8.

[Sut11] A. V. Sutherland, Computing Hilbert class polynomials with the Chinese remainder theorem, *Math. Comp.* 80 (2011), 501–538, doi:10.1090/S0025-5718-2010-02373-7.

[Sut12] A. V. Sutherland, Accelerating the CM method, *LMS J. Comput. Math.* 15 (2012), 172–204, doi:10.1112/S1461157012001015.

[Sut13] A. V. Sutherland, Isogeny volcanoes, in *ANTS-X*, Open Book Ser. 1, MSP, 2013, 507–530, doi:10.2140/obs.2013.1.507.

[t5k] The PrimePages, Top twenty: Elliptic Curve Primality Proof, `https://t5k.org/top20/page.php?id=27` (accessed Aug 2026).

[Wat69] W. C. Waterhouse, Abelian varieties over finite fields, *Ann. Sci. École Norm. Sup.* (4) 2 (1969), 521–560, doi:10.24033/asens.1183.

[DANGER3] A. V. Sutherland, *DANGER3* (Pomerance triples: definition, verifier `vpp.py`, Lean verifier, exhaustive data to 2^{32}, challenge log), `https://github.com/AndrewVSutherland/DANGER3`, 2026 (commit `3cd7be6`, accessed August 19, 2026).

[OneShot] A. V. Sutherland, *OneShotPrimalityProofs* (one-shot ECPP: definition, verifier `voneshot.py`, finder `oneshot.gp`, exhaustive data to 2^8, challenge table), `https://github.com/AndrewVSutherland/OneShotPrimalityProofs`, 2026 (commit `188f53e`, accessed August 19, 2026). Part of the DARPA expMath program.

[Short] A. V. Sutherland, *ShortPrimalityProofs* (short ECPP: definition, verifier `vsmallECPP.py`, finders `short.gp`/`parallel_short.py`, exhaustive data to 2^8, challenge table), `https://github.com/AndrewVSutherland/ShortPrimalityProofs`, 2026 (commit `0408b2b`, accessed August 19, 2026).

[OSF] A. V. Sutherland and Claude Code, *OneShotFastECPP* (the CM one-shot engine: `dscan`/`smooth`/`fproot`/`cm_method`/`oneshot`; reports *descent-chain*, *sea-crossover*, and this write-up; verified certificates), `https://github.com/AndrewVSutherland2/OneShotFastECPP`, 2026 (branch `short-ecpp`).